



Simulation of neutron guides using McStas

ISIS McStas School 2025

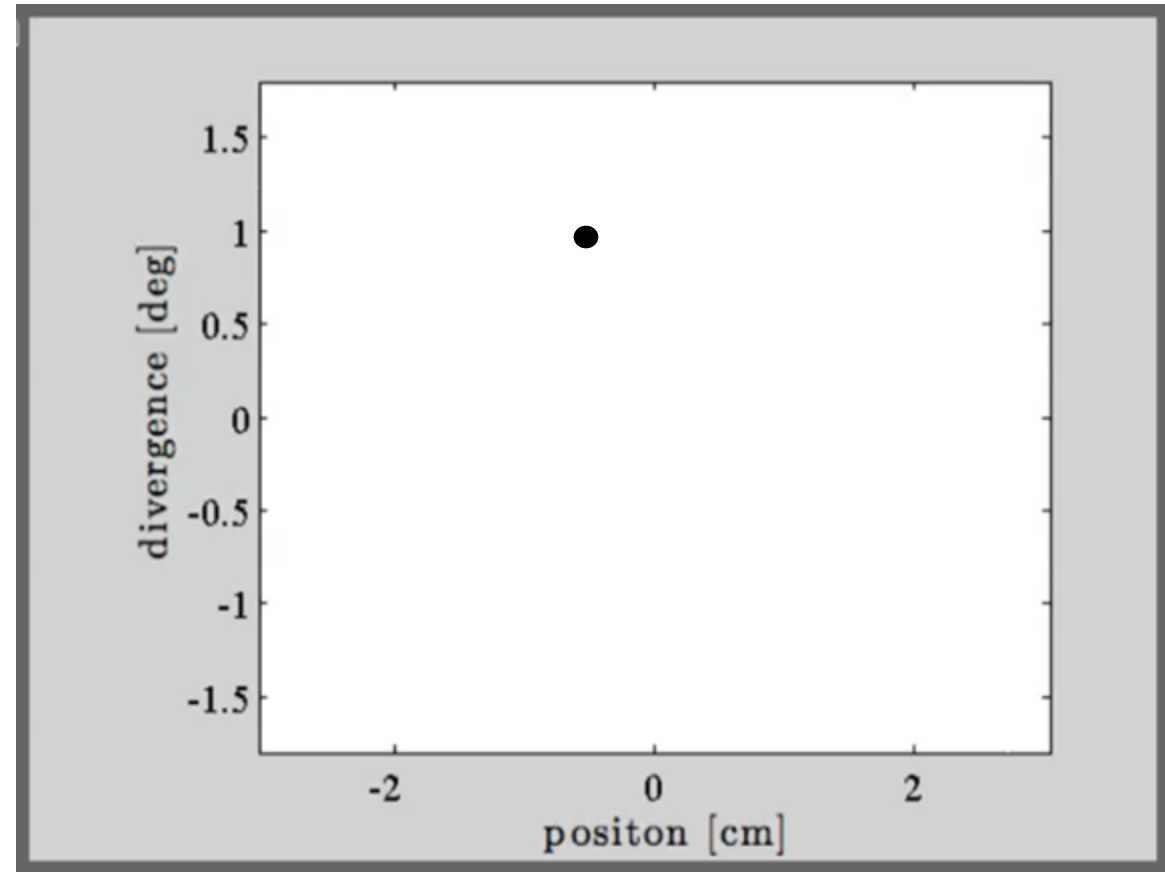
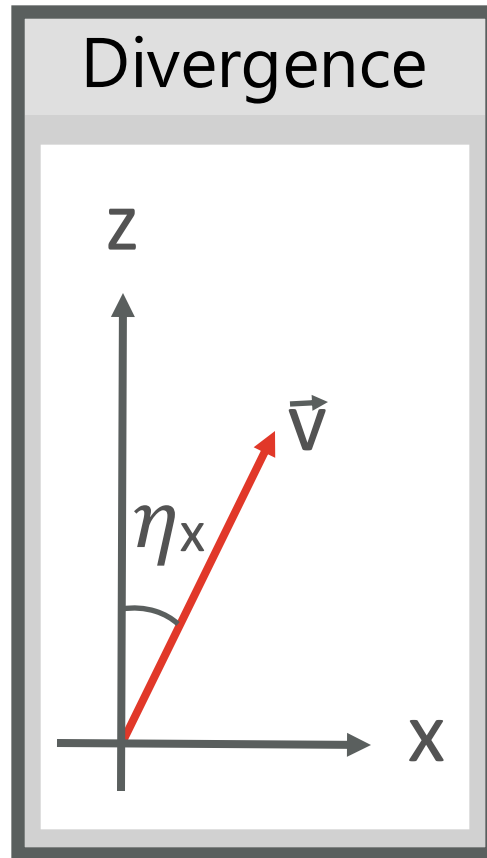
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Data Management and Software Centre
European Spallation Source

2025-11-04

1. A beam in free space
2. Neutron mirrors
3. Guides
4. Designing guides
5. Simulation
6. Workflow

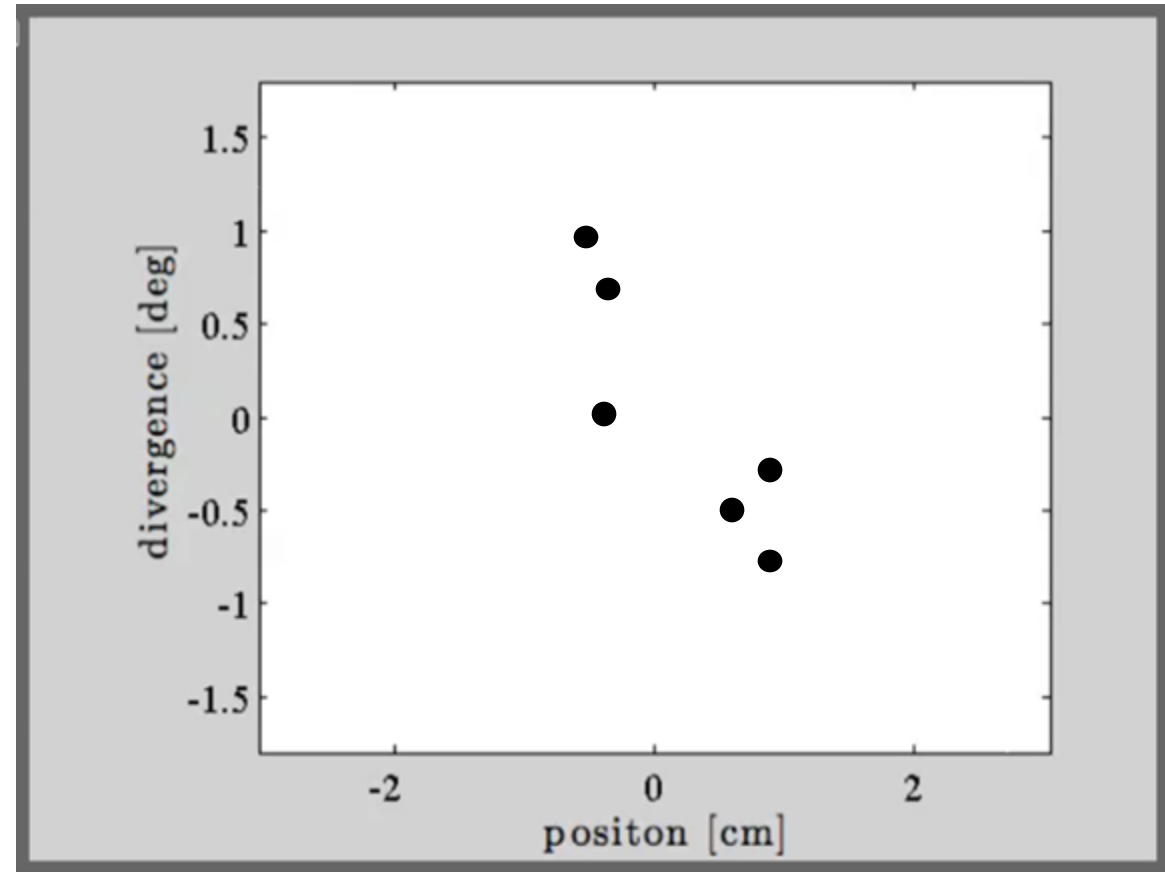
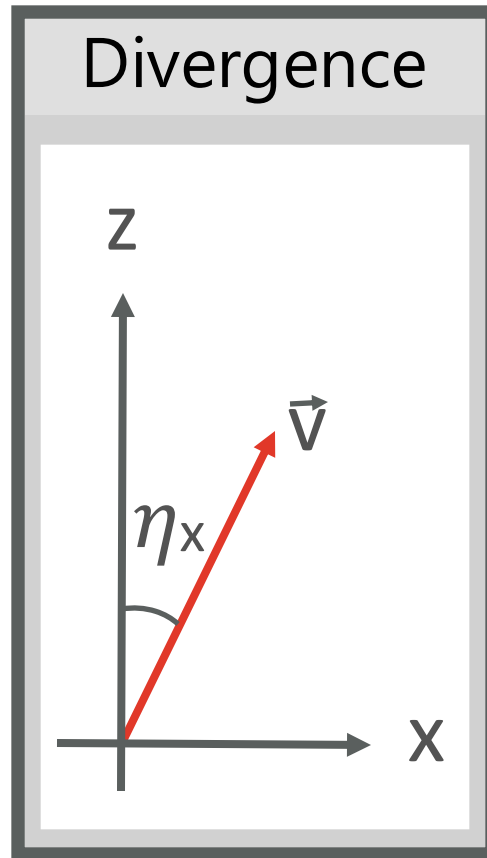
Introduction

Beam and its propagation



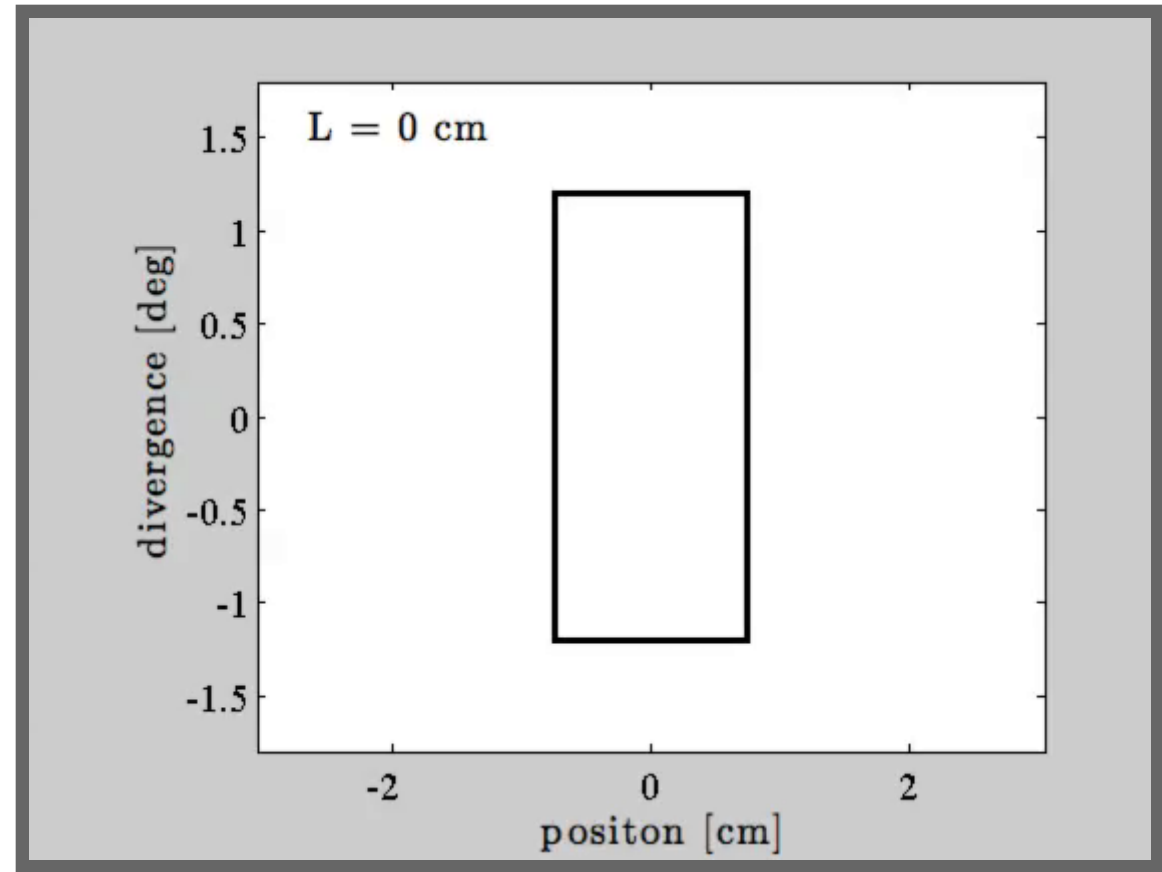
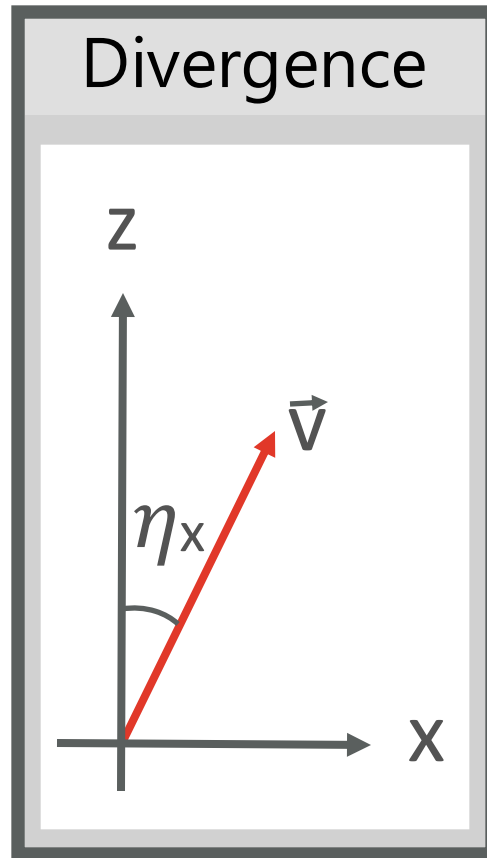
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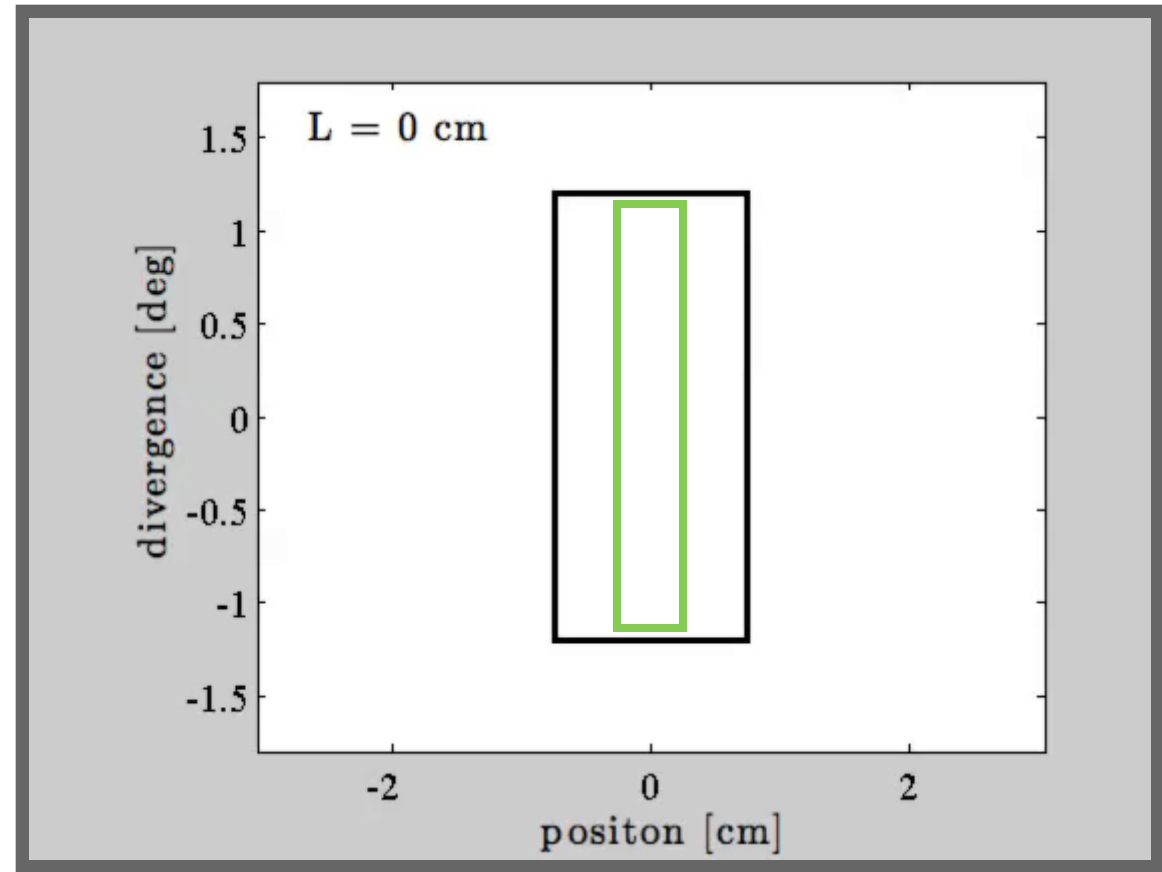
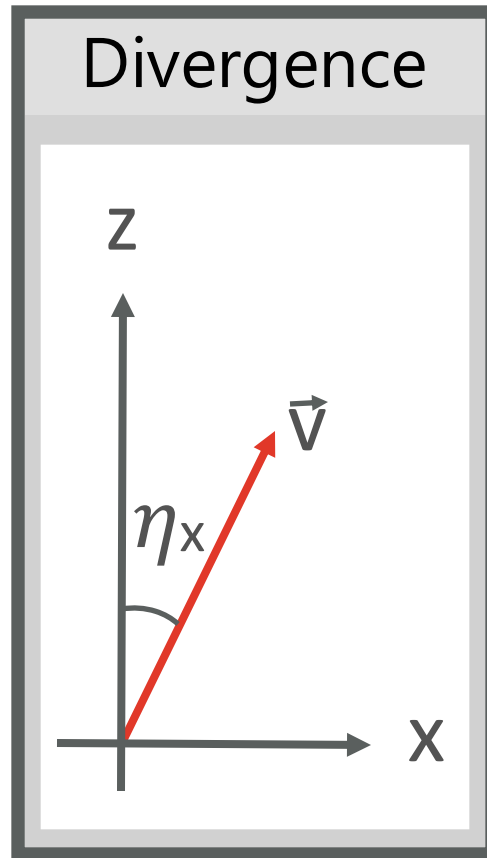
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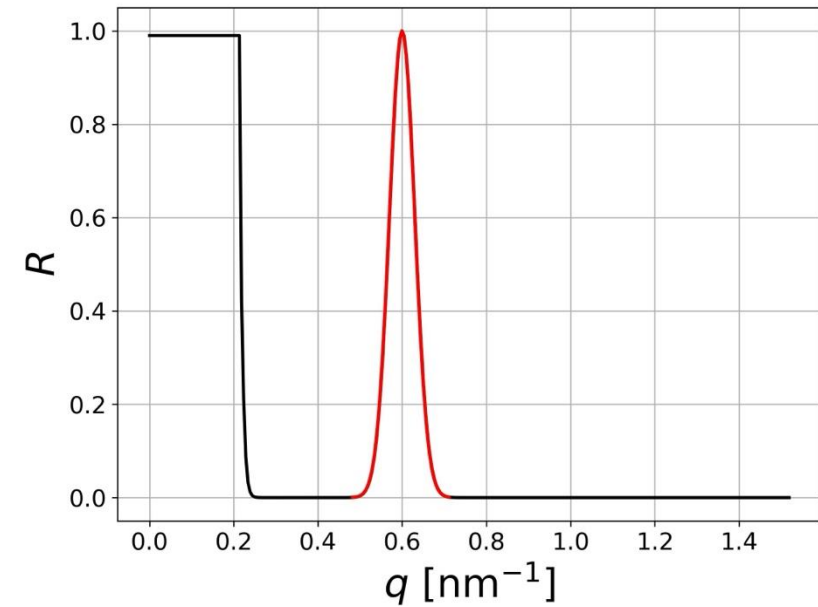
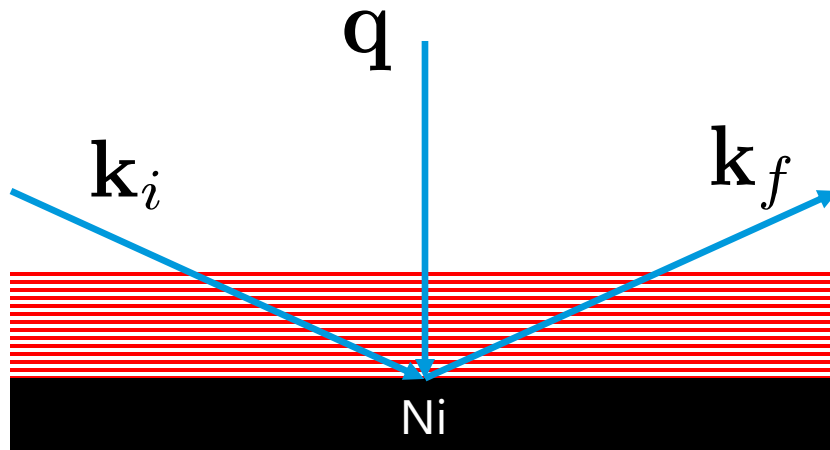
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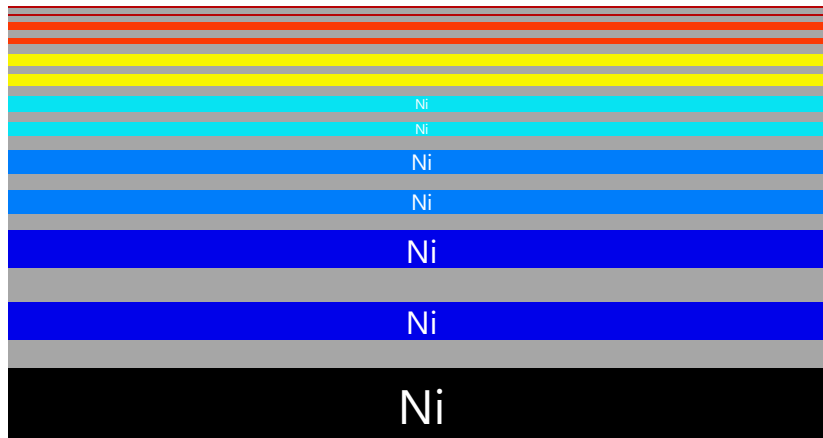
Introduction

Neutron mirrors

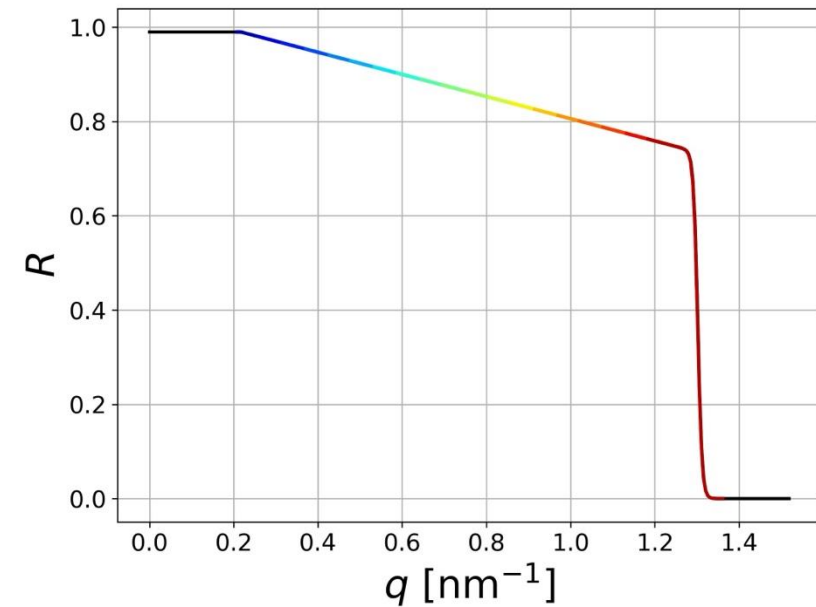


Introduction

Neutron super-mirrors

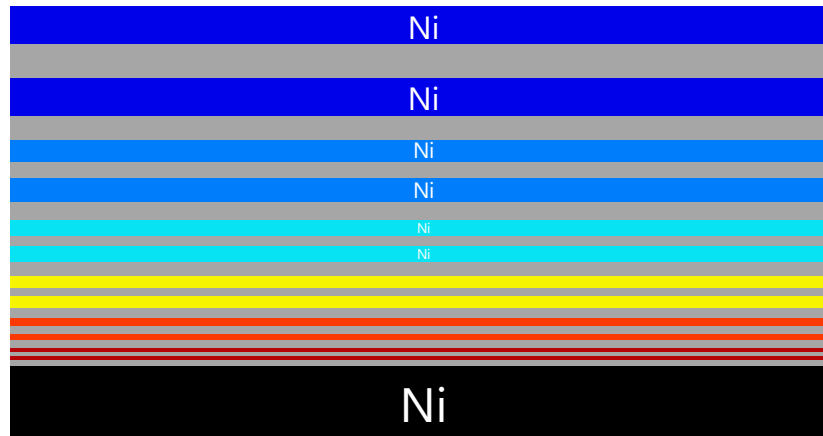


Ti as spacing for high contrast

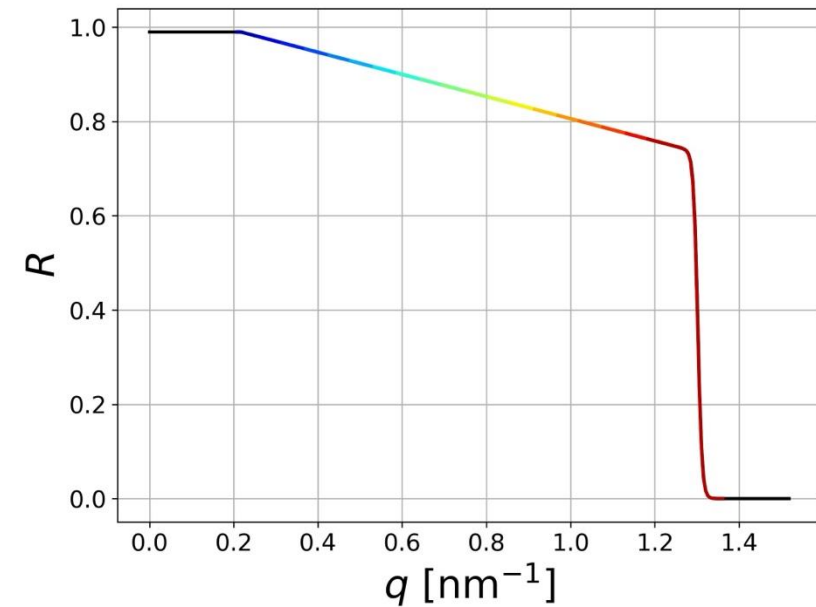


Introduction

Neutron super-mirrors



Ti as spacing for high contrast



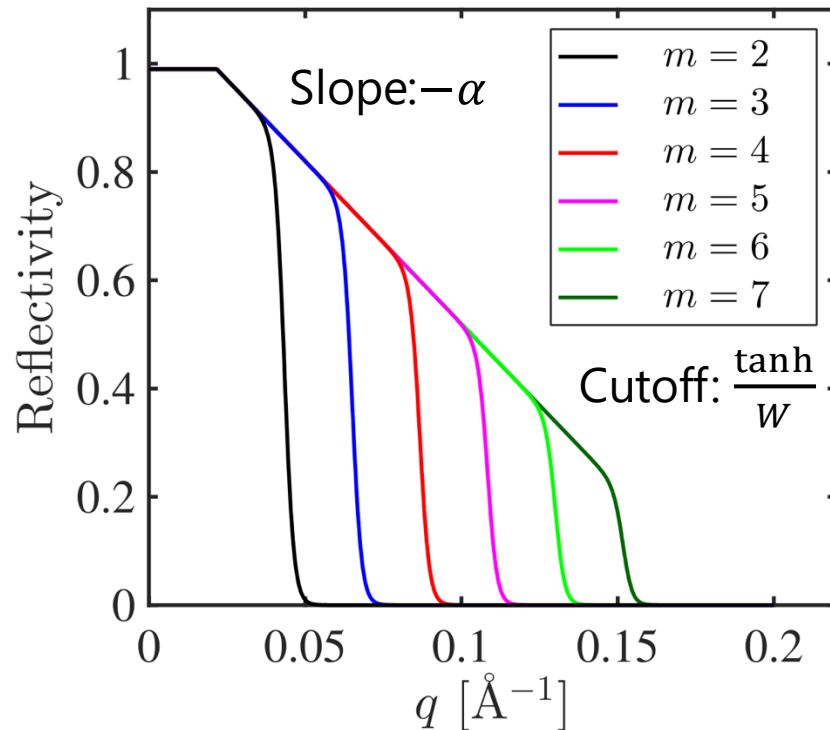
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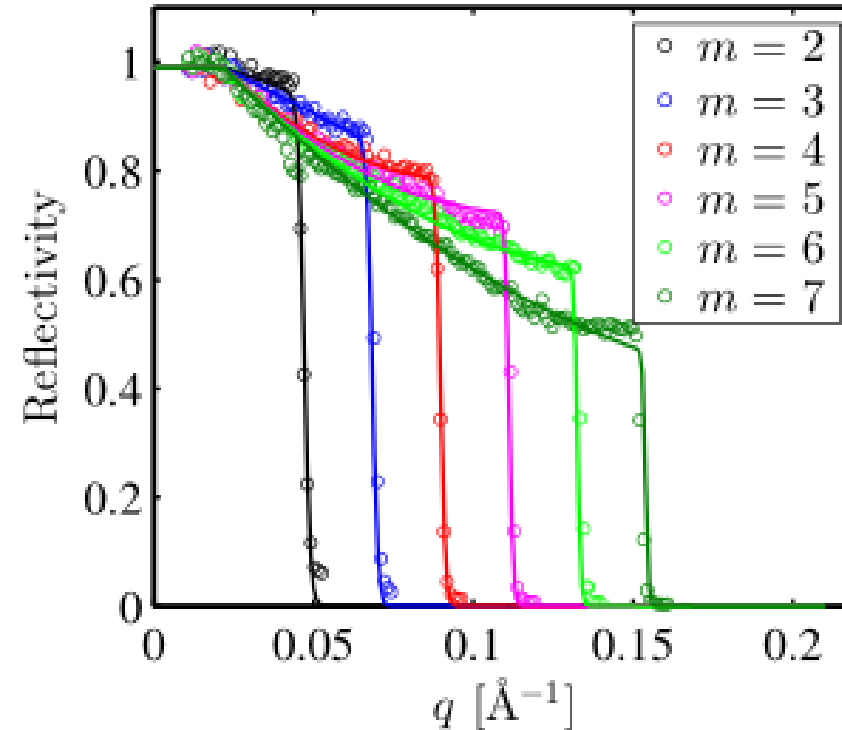


$$R(q) = \begin{cases} R_0 & \text{if } q < q_c \\ R_0(1 - \tanh((q - mq_c)/W))(1 - \alpha(q - q_c))/2 & \text{otherwise} \end{cases}$$

McStas standard model



McStas fitted model



$\alpha = 0$
 $W = 0$
Only m
matters
Better mirrors
available
today

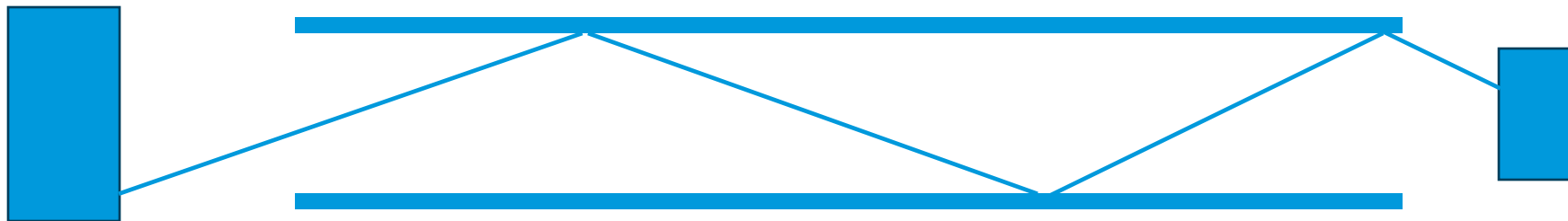
Introduction

Neutron guide

Neutron source

Guide

Destination



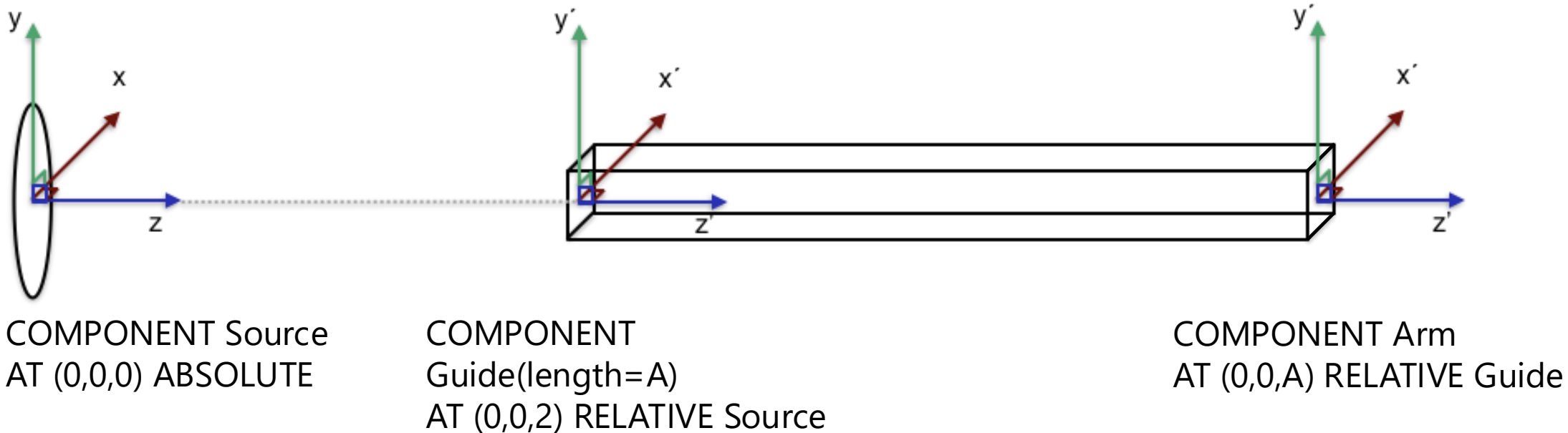
Introduction

Guide placement in McStas



The center is the front of the guide element

Tip: Insert a guide at the end of the guide



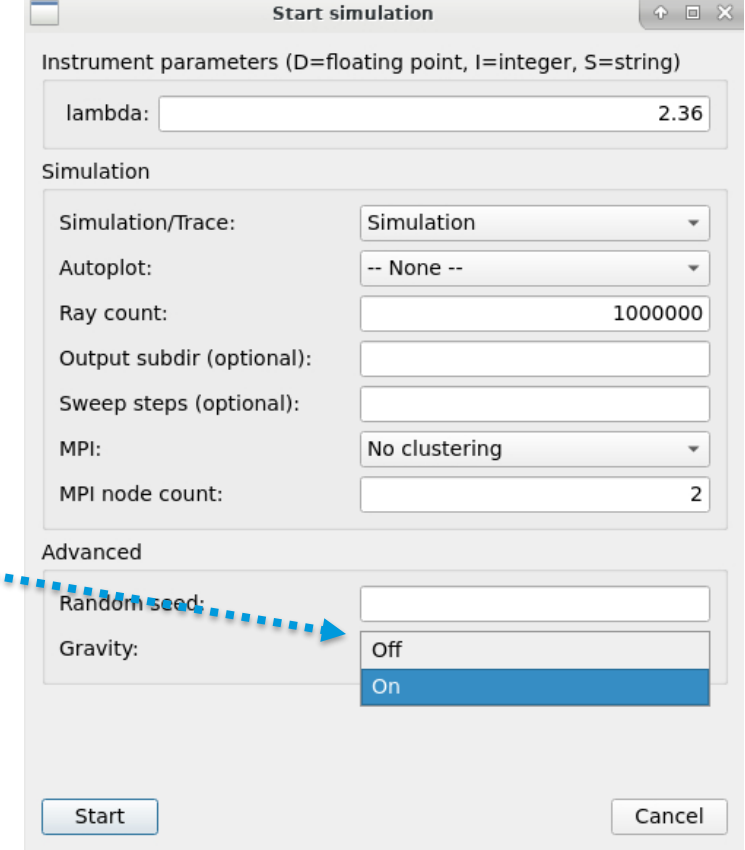
Introduction

Gravity in McStas

- Enabled by adding `-g / --gravitation` on command line or by selecting "Gravity On" in mcgui
- Default ~ gravity on earth

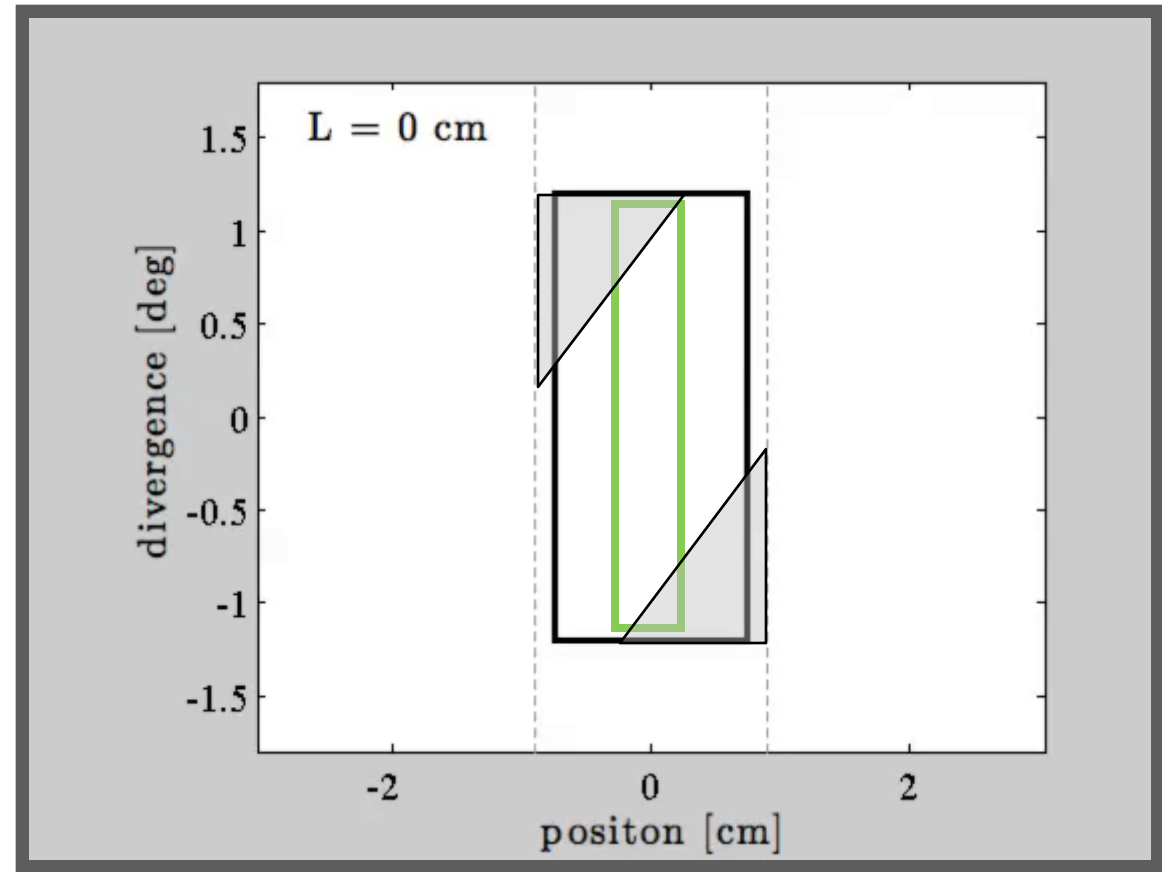
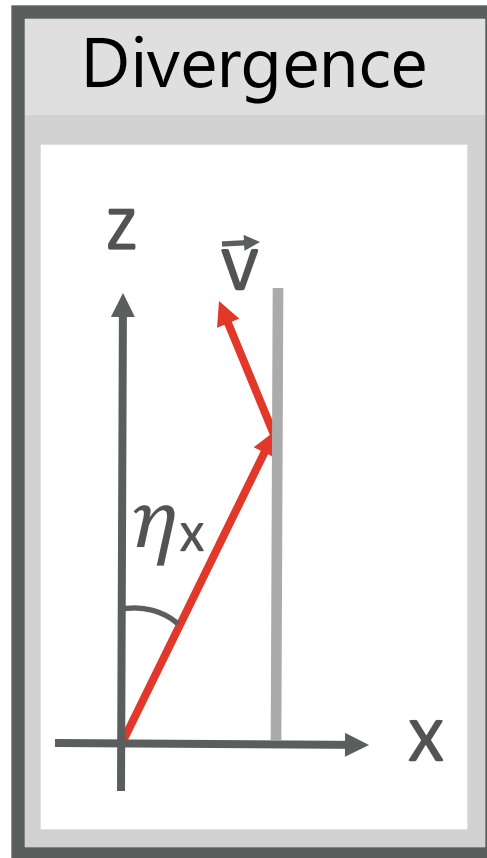
```
#define GRAVITY 9.81 /* [m/s^2] gravitational acceleration */
```


(If on the moon, use `-DGRAVITY=1.62 ;-`)
- For guides, only `Guide_gravity` and `Elliptic_guide_gravity` support parabolic propagation. (Many others propagate linearly in $\vec{v}$ direction.)
- As you will see in the practical, implications are greatest with long wavelengths and at long distances
- "How about e.g. elliptic mirror optic X that does not support gravity?"
 - often a good workaround is to add a monitor close to the surface of object X, this takes care that propagation up to the monitor includes gravitation:
 - Gravity is enabled in any call to `PROP_DT`, `PROP_Z0` etc., but not in `intersect_*` routines (most monitors use `PROP_Z0` directly, no `intersect_*` call first)
 - OK to propagate without gravitation e.g. within sample, through velocity selector etc. / range of ~cm's



Introduction

Neutron guide in phase-space

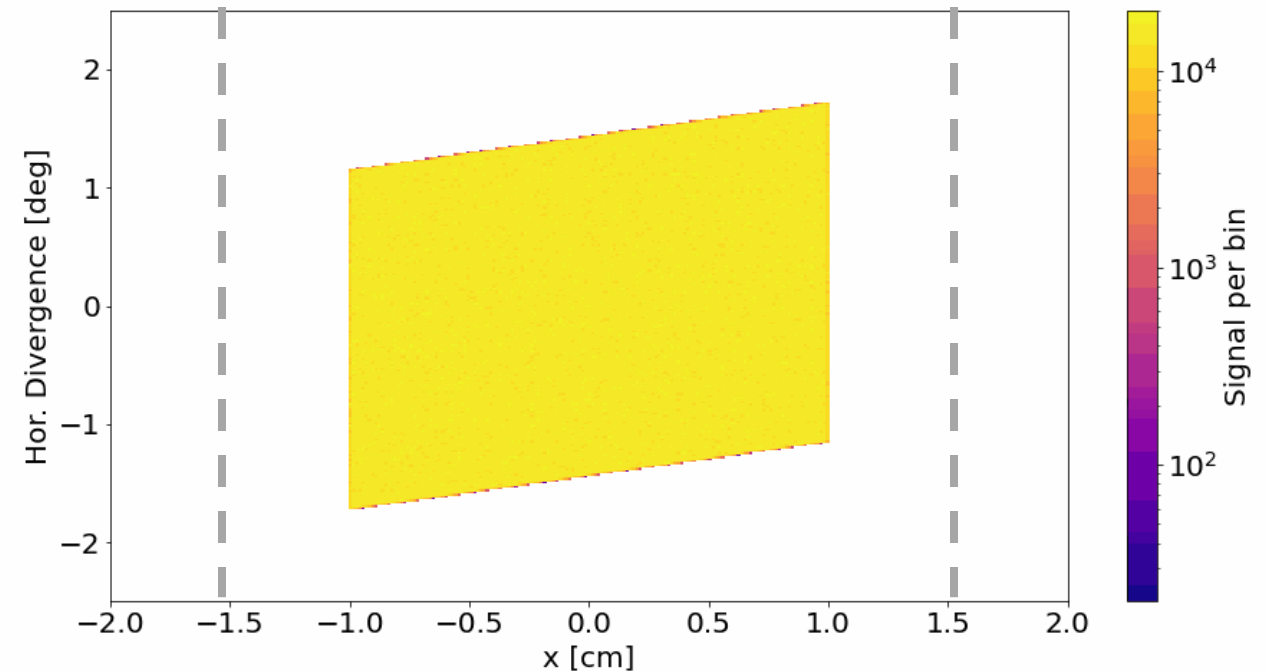


Introduction

Phase-space theory



- Phase-space properties
 - Often 3D $\rightarrow$ 2 x 2D
 - Propagation (no paths crossing)
 - Liouville's theorem
 - phase-space density can not be increased
- Brilliance transfer
- Losses from:
 - Reflectivity less than 1
 - Not full coverage of phase-space



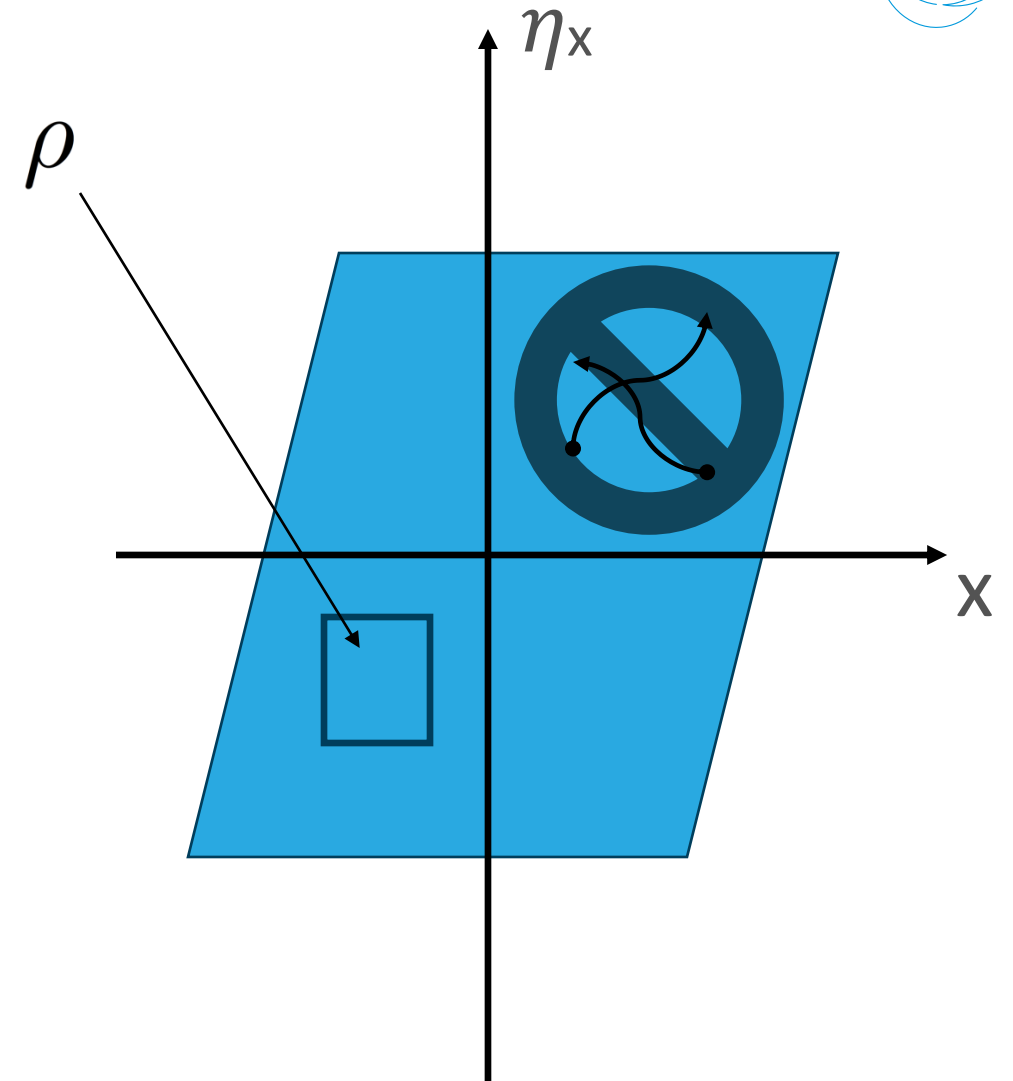
Introduction

Phase-space theory

- Phase-space properties
 - Position & velocity $\rightarrow x, \eta_x, y, \eta_y, \lambda$
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$$\frac{d\rho}{dt} \leq 0$$

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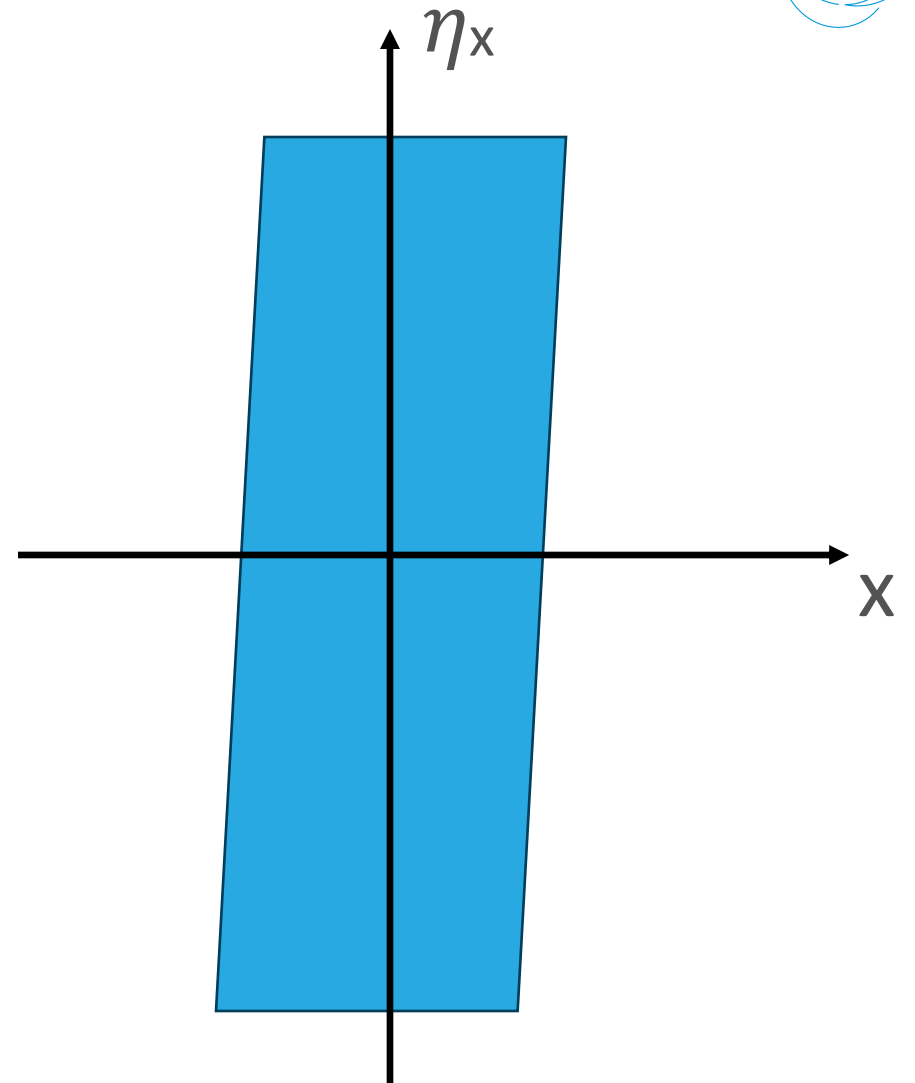
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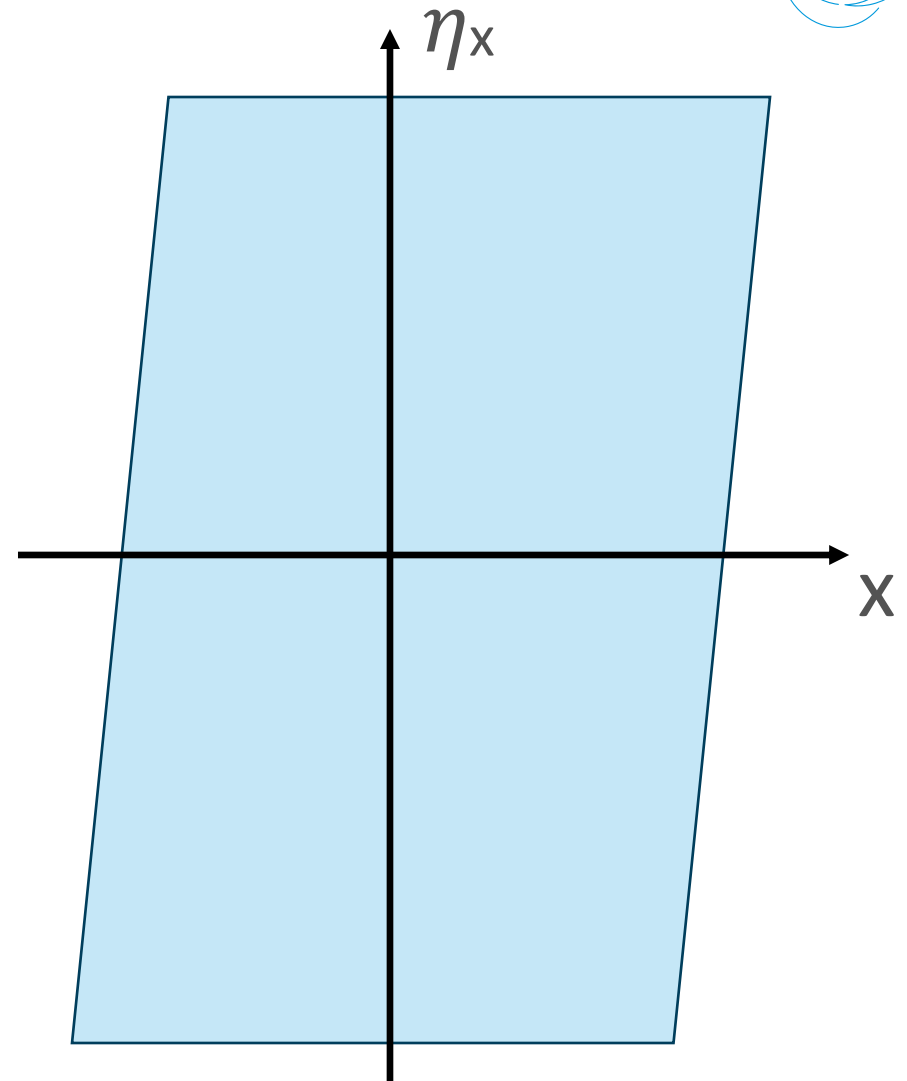
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Phase-space theory

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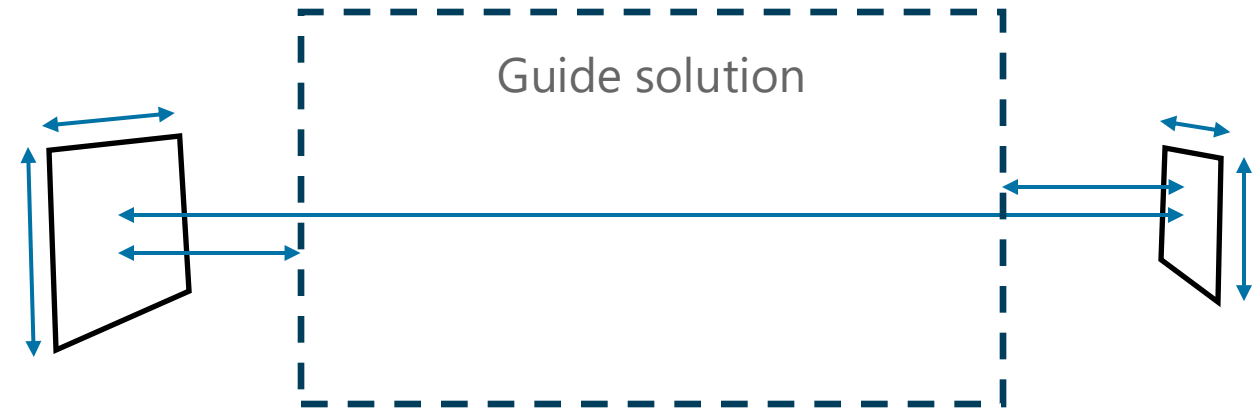


Introduction

Define the problem of guide design



- Source
 - Dimensions
- Space constraints
 - Source to optics
 - Optics to destination
- Desired beam
 - Dimensions
 - Divergence
 - Wavelength interval
 - Distance from source
- At least 11 important parameters



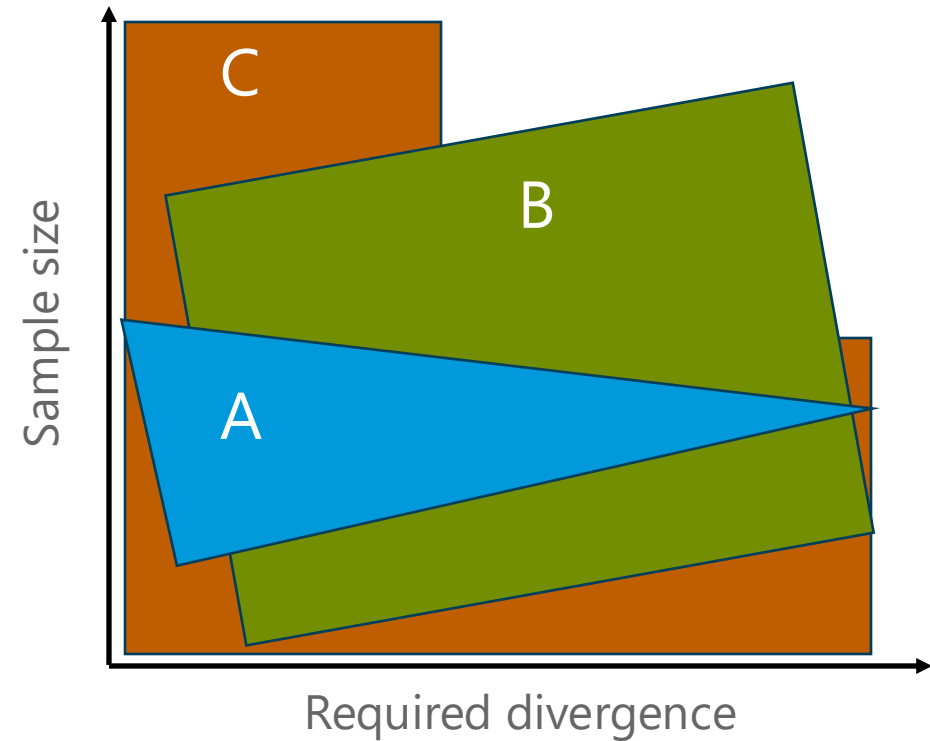
In addition:

- Constraints for width and height
- Required gaps for equipment
- Breaking line of sight to reduce background

Introduction

Overall problem

- Not solved
- No solution map from requirements to guide design
- Recommendation map, maybe
- Need to evaluate different possible solutions and compare



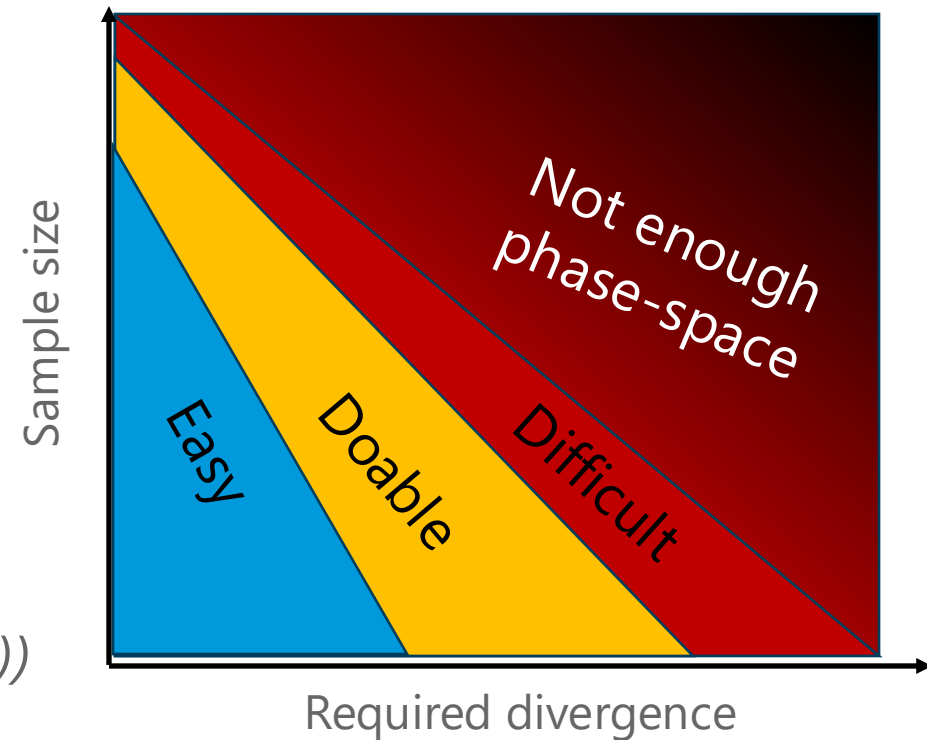
Introduction

Overall problem

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Focusing case: $WM/L_M \geq 2\phi S$
Non-focusing case: $WM/L_M \geq 2\phi(S + 2L_S \tan(\phi))$

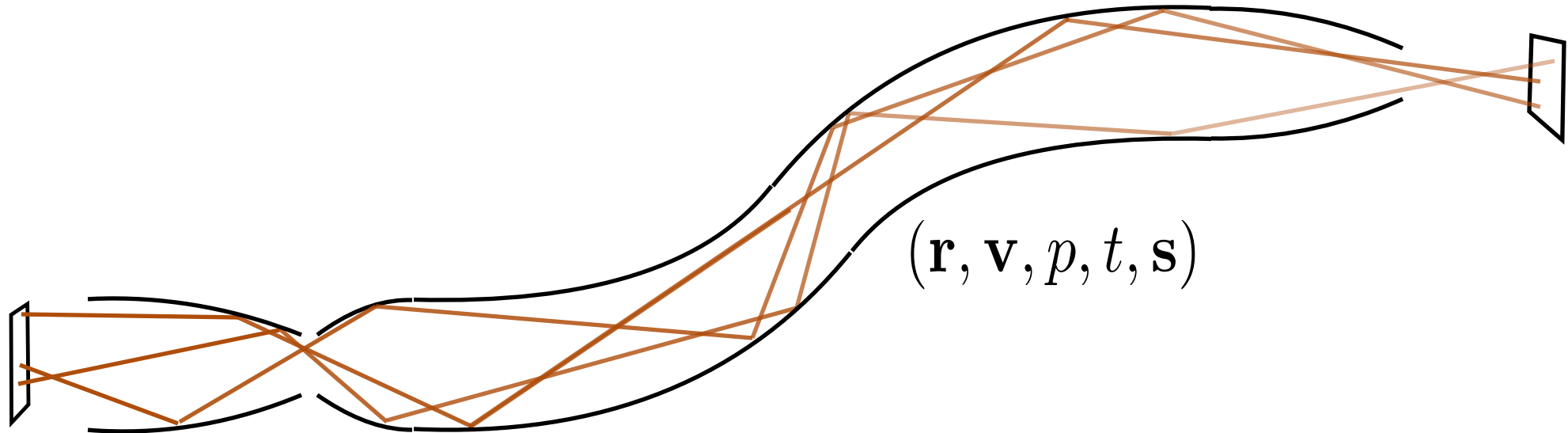
W : Guide size M : Moderator size L_M : Guide moderator distance
 ϕ : Divergence requirement S : Sample size L_S : Sample distance



Simulation

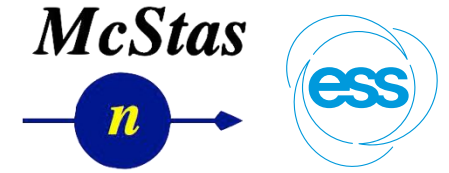
Evaluate a design

- Monte Carlo ray-tracing for evaluating performance
- Propagate many rays through optical system

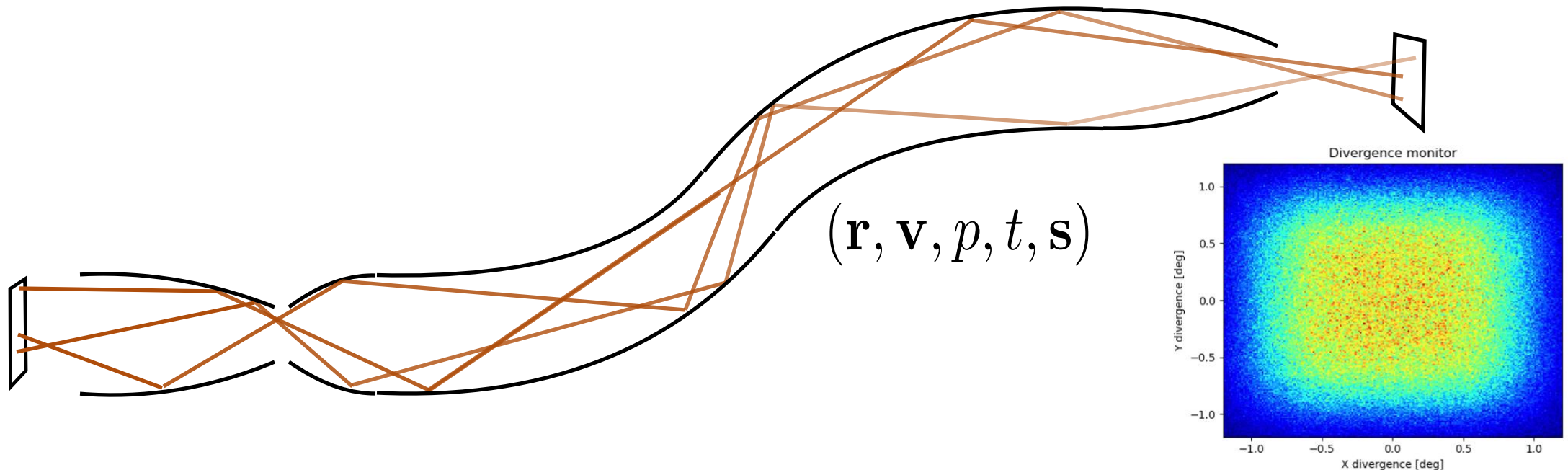


Simulation

Evaluate a design

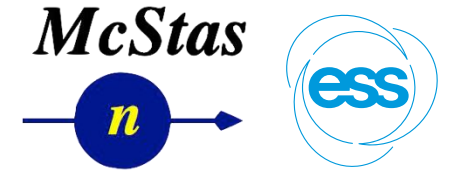


- Monte Carlo ray-tracing for evaluating performance
- Propagate many rays through optical system
- McStas has large library of components
- Community project, components contributed



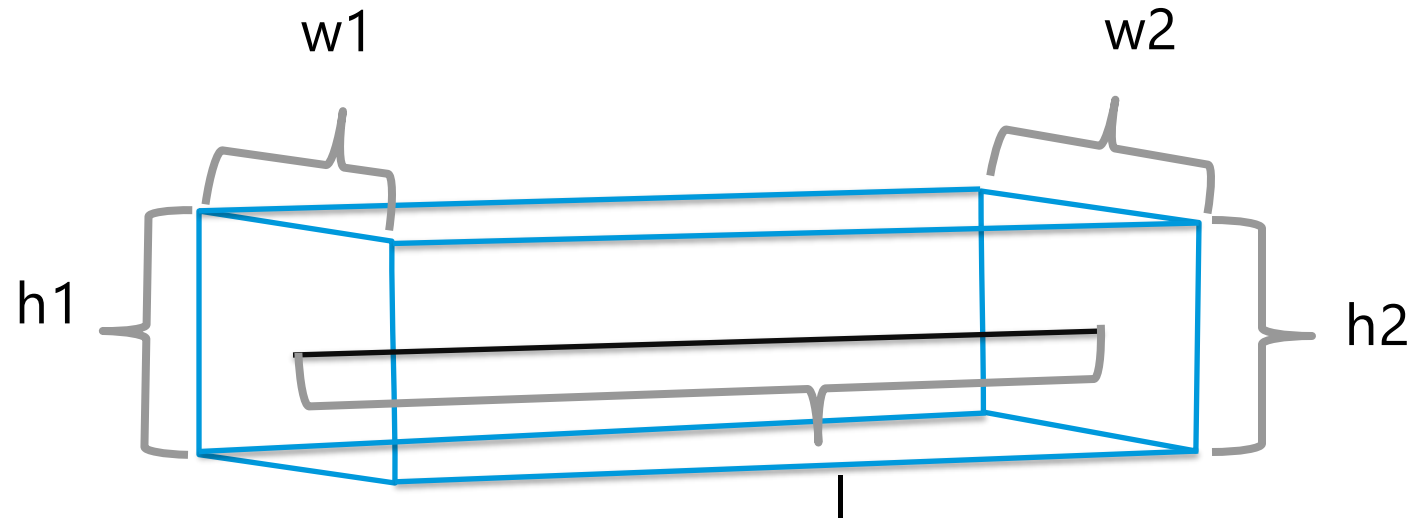
Simulation

Popular components: **Guide_gravity**



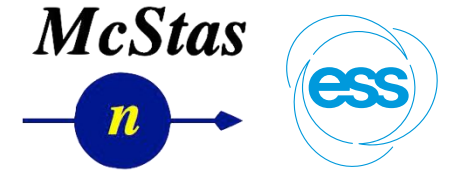
Important guide component with gravity, parameter-interface similar to e.g. Guide.comp

Many additional features, channels, fermi chopper, ... (see mcdoc pages for more info)



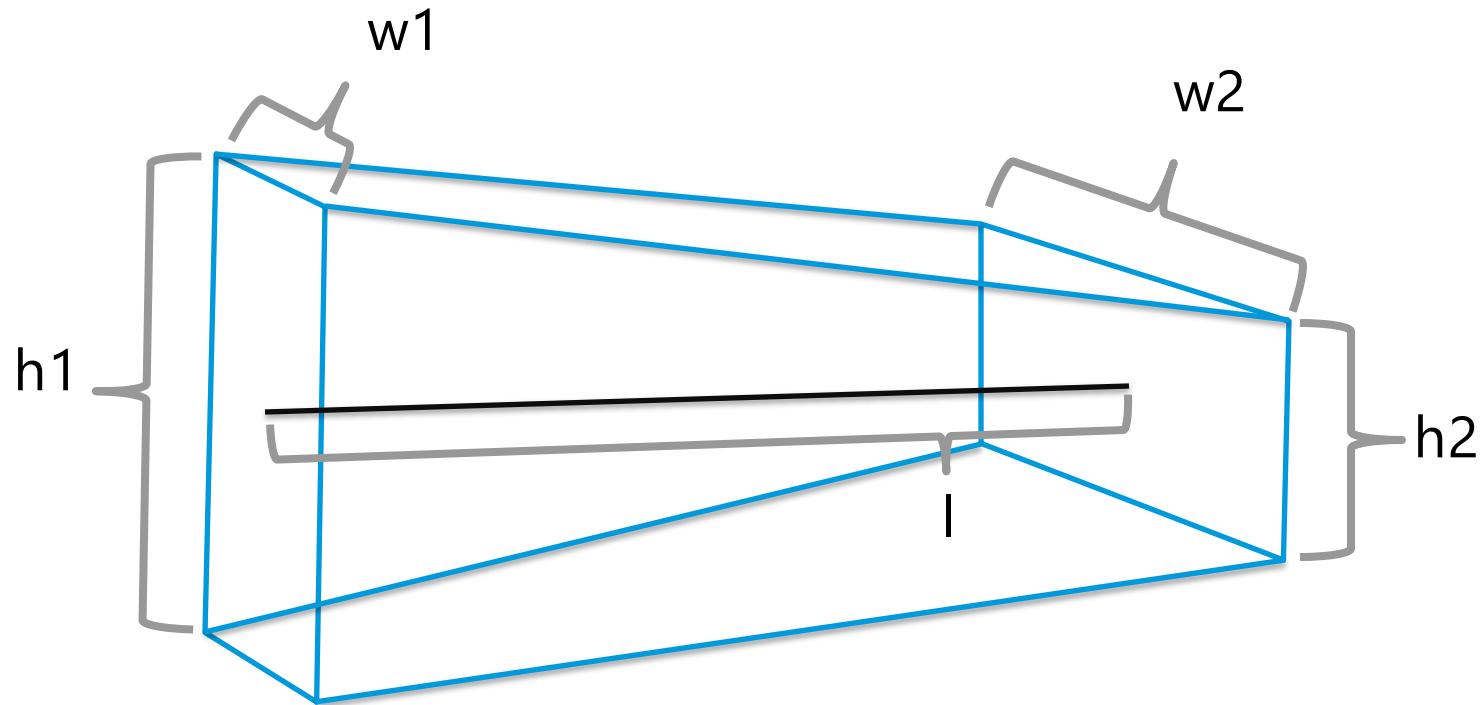
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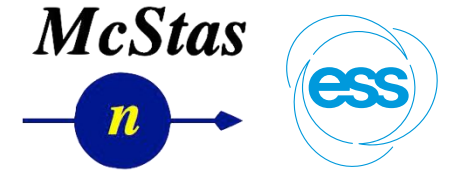
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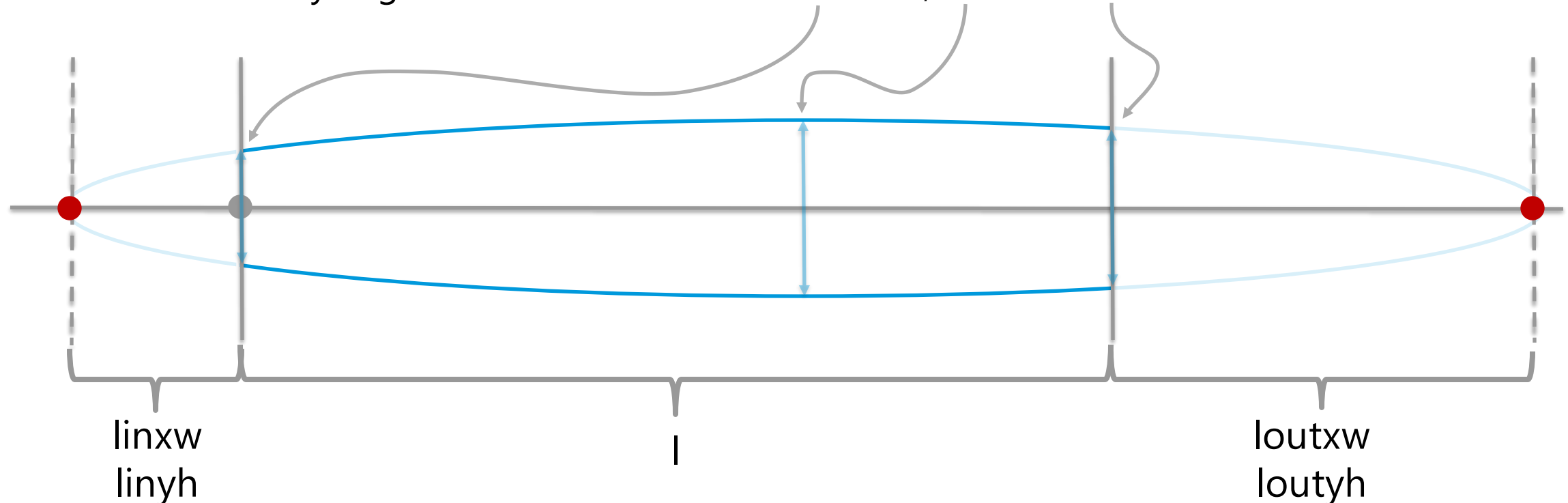
Simulation

Popular components: **Elliptic_guide_gravity**



Useful for elliptic and parabolic guide geometries, focusing, ballistic, coating distribution, ...

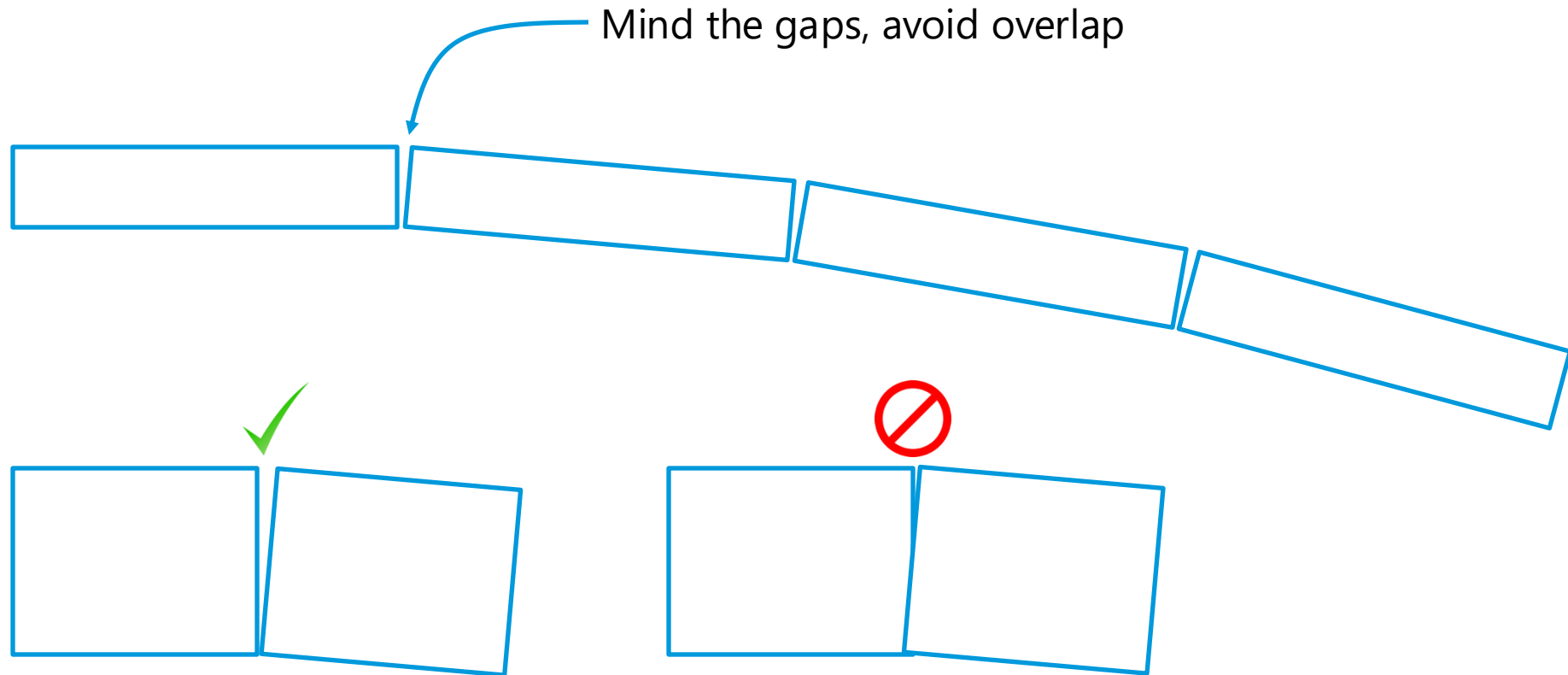
xwidth and yheight at DimensionsAt = "entrance" , "mid" or "exit"



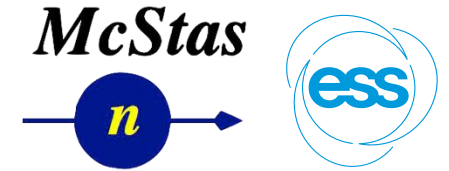
Simulation

Breaking line of sight

Bender / Guide_curved component or many straight sections



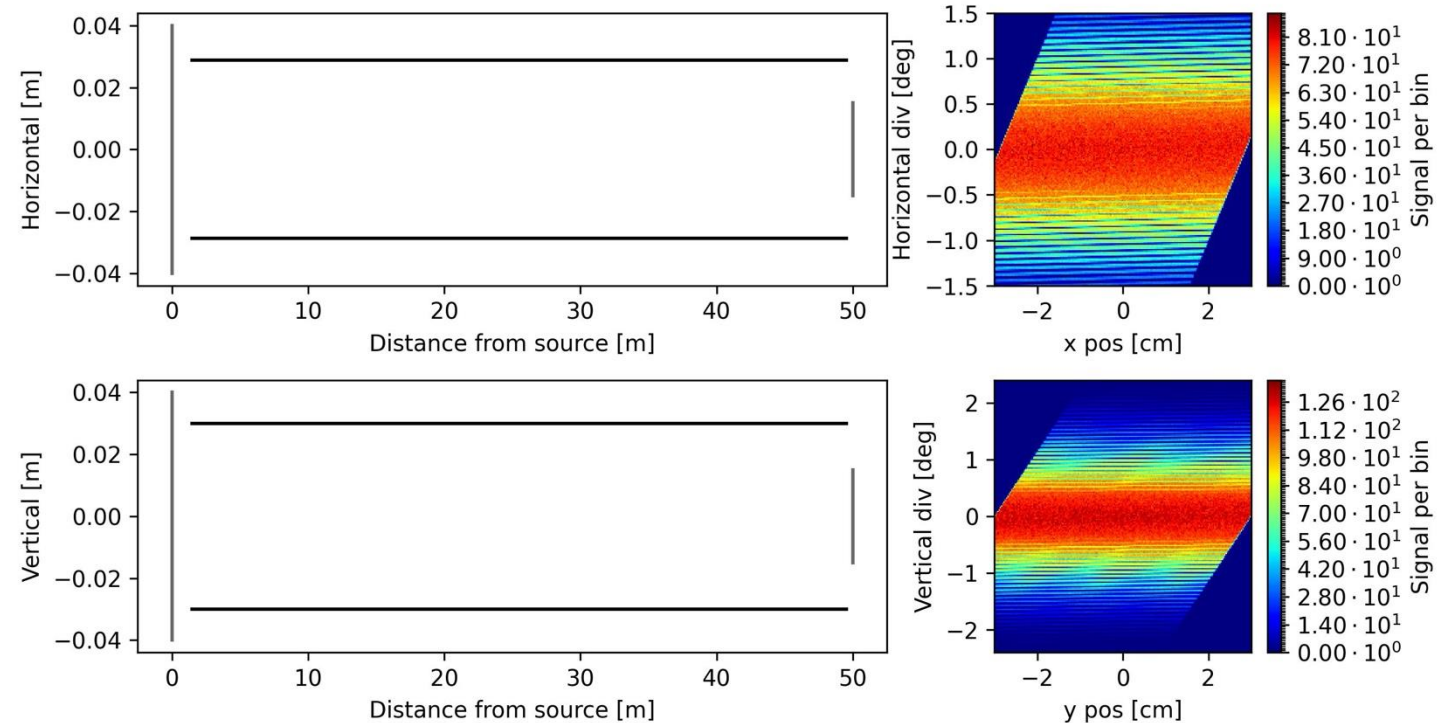
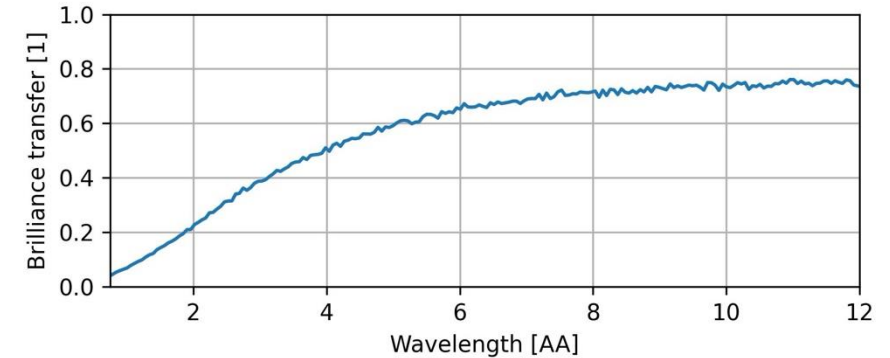
Solutions / Philosophies / Paradigms



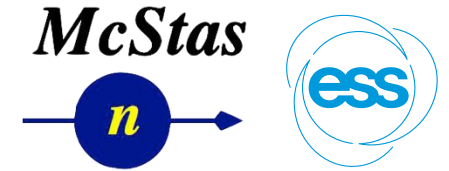
Traditional guides

- Rectangular cross section
- Sensitive to distance to sample
- Handles low divergence well

$3 \times 3 \text{ cm}^2$
 $\pm 0.75^\circ \text{ H}$
 $\pm 1.20^\circ \text{ V}$



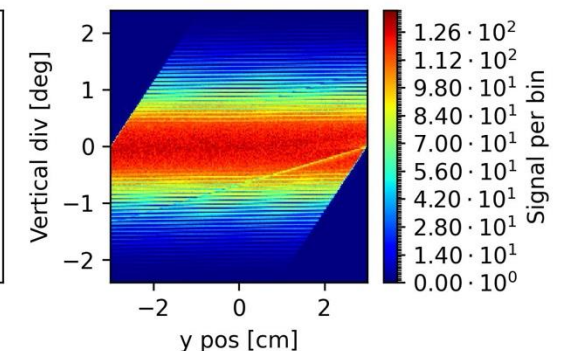
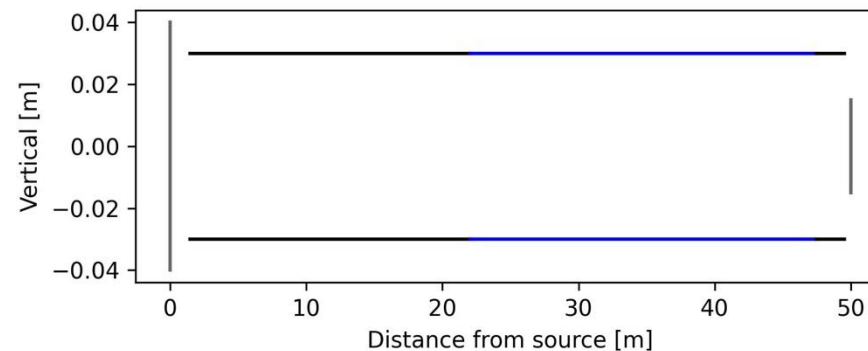
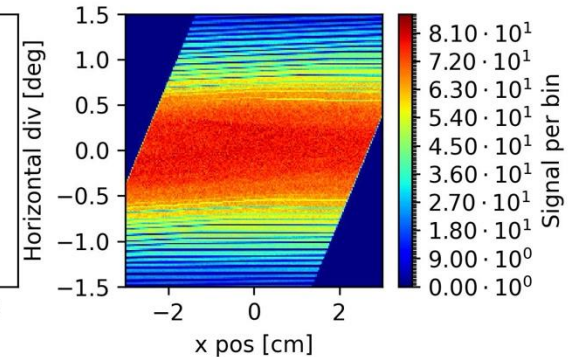
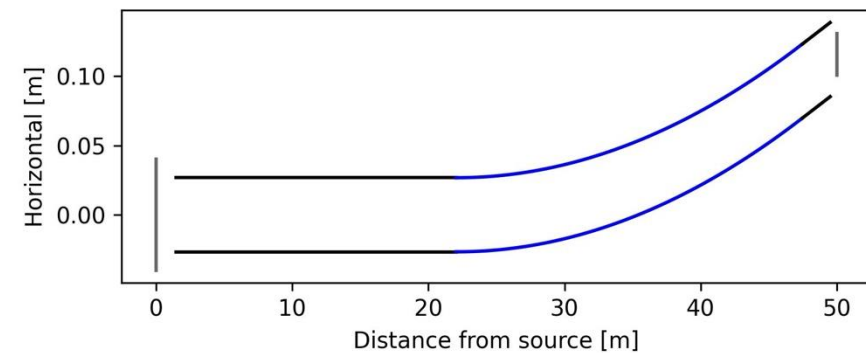
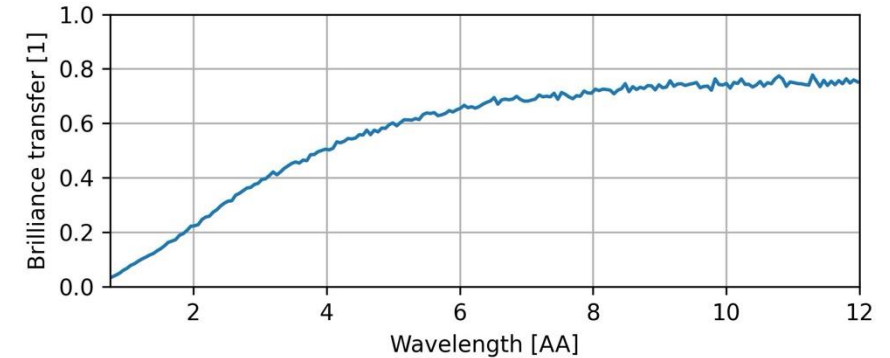
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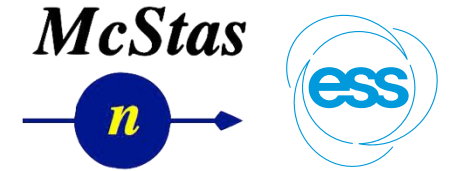
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- Rectangular cross section
- Sensitive to distance to sample
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- Curve out of line of sight

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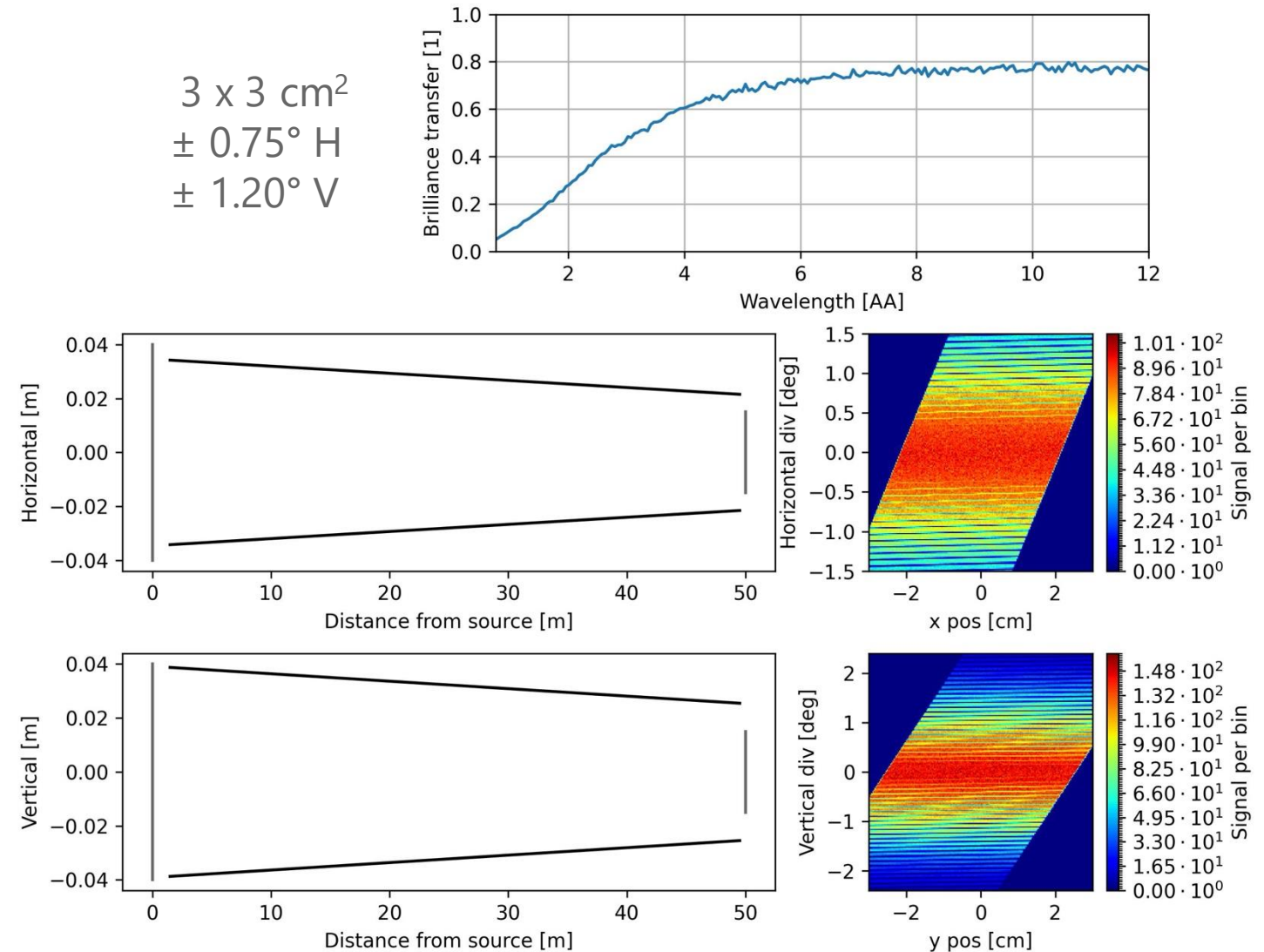


Solutions / Philosophies / Paradigms

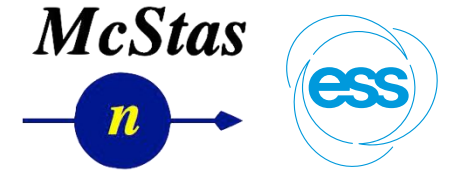


Traditional guides

- Rectangular cross section
- Sensitive to distance to sample
- Handles low divergence well
- Curve out of line of sight
- Can be tapered - artifacts



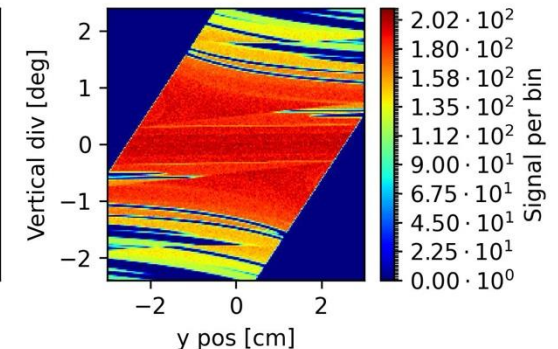
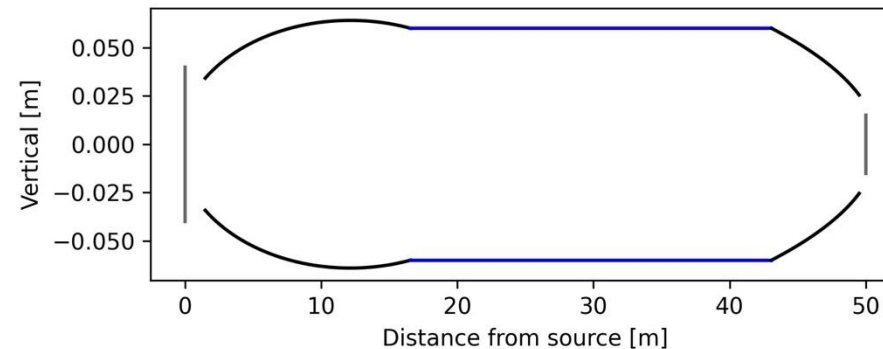
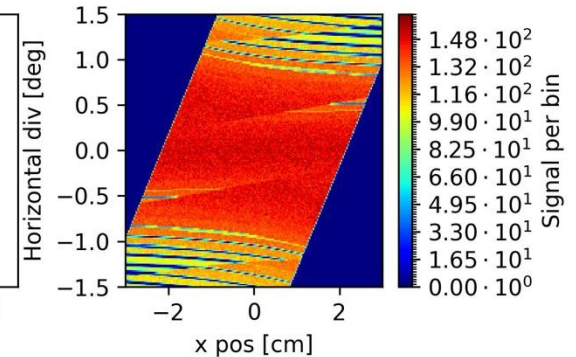
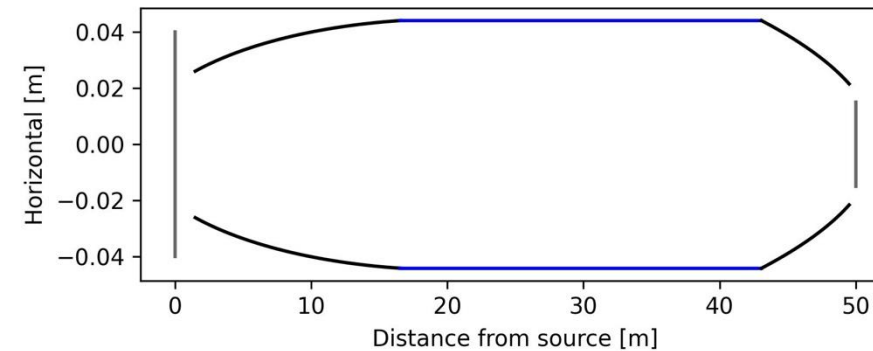
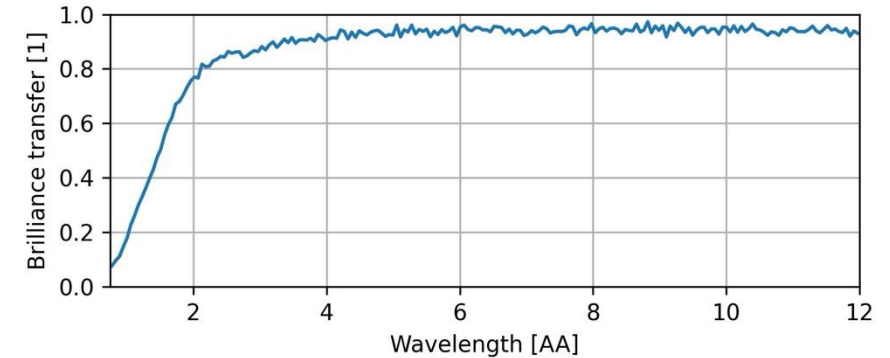
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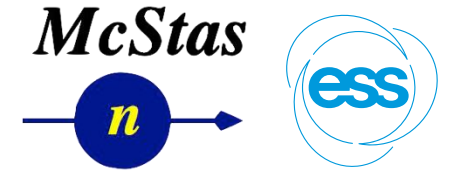
Ballistic guides

- Increase and decrease rectangular cross section
 - Reduce divergence for transport
 - Increase divergence for delivery
- Sensitive to distance to sample
- Handles larger divergence

$3 \times 3 \text{ cm}^2$
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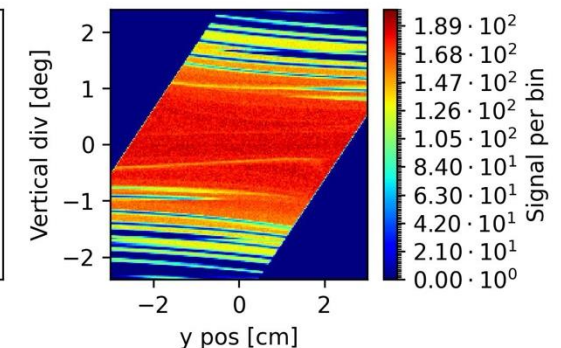
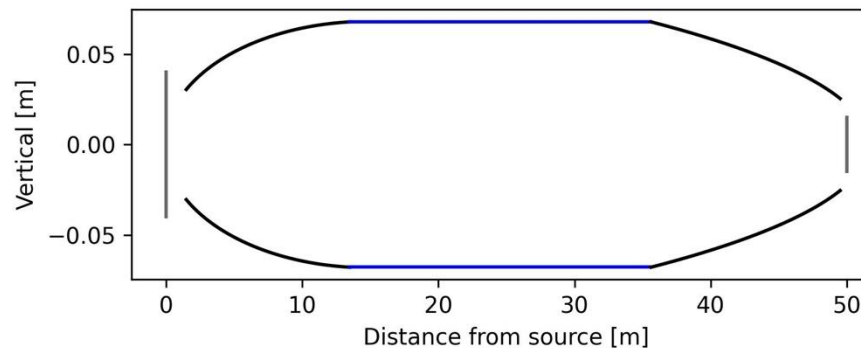
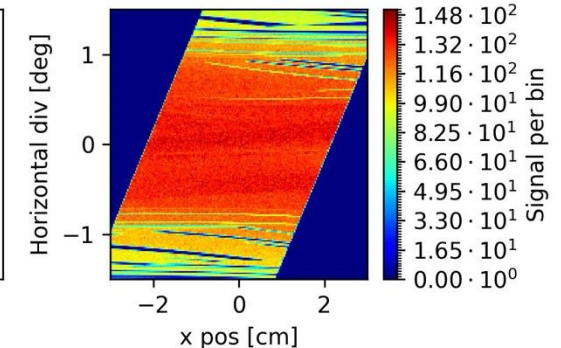
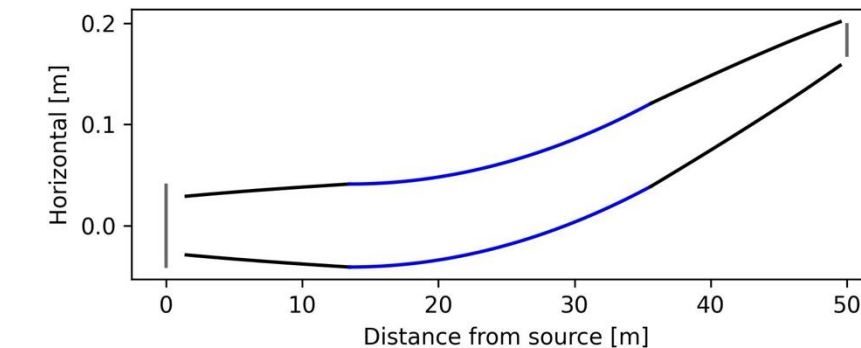
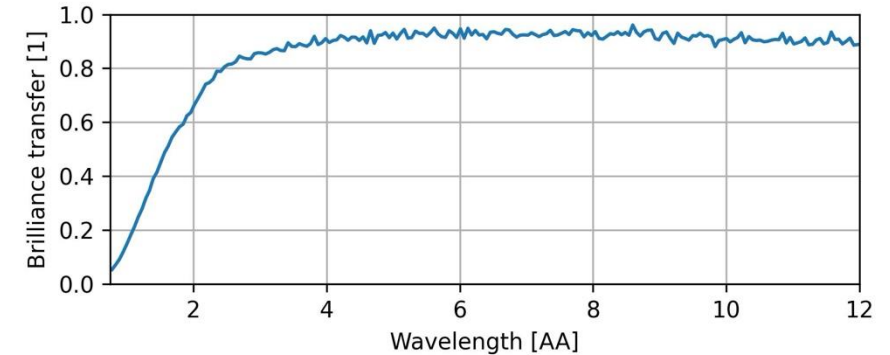
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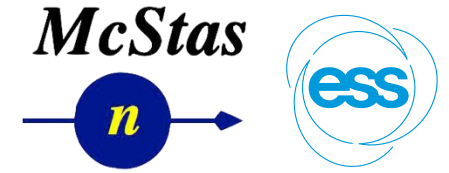
Ballistic guides

- Increase and decrease rectangular cross section
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 - Increase divergence for delivery
- Sensitive to distance to sample
- Handles larger divergence
- Can be curved

$3 \times 3 \text{ cm}^2$
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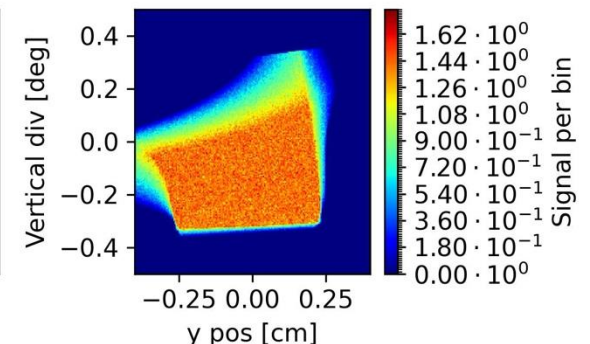
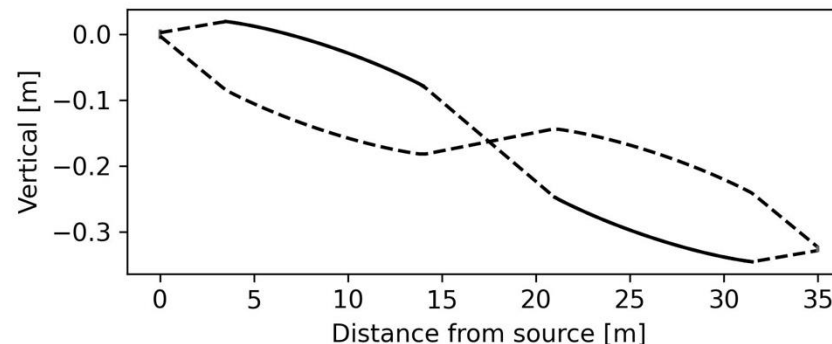
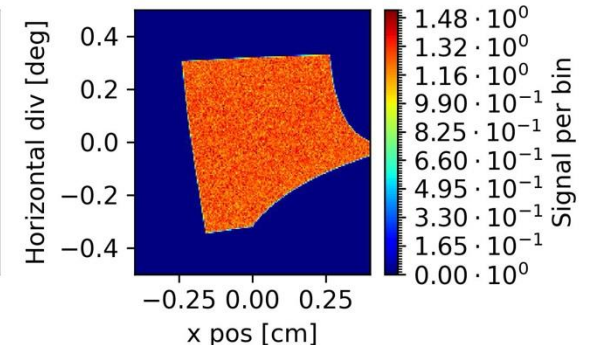
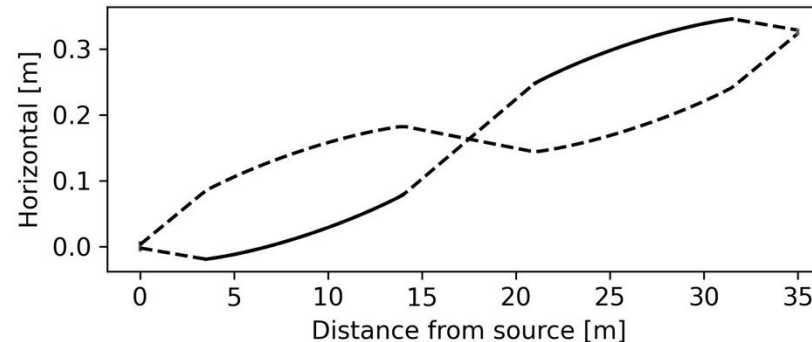
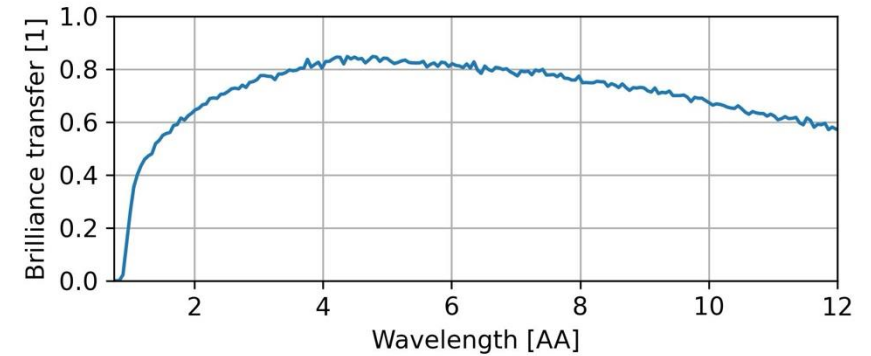
Solutions / Philosophies / Paradigms



Selene – Montel optics

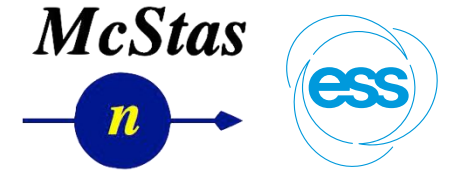
- Focusing design
- Allows large distance to sample
- Point to point, scalable, analytical calculation
- Required coating calculated
- Provides control over beam
- Limits:
 - Gravity
 - Beamsize – length connection

$4 \times 4 \text{ mm}^2$
 $\pm 0.25^\circ \text{ H}$
 $\pm 0.25^\circ \text{ V}$



J. Stahn, T. Panzner, U. Filges, C. Marcelot, P. Böni
Study on a focusing, low-background neutron delivery system
NIMA, **2011**, Volume 634, page S12-S16

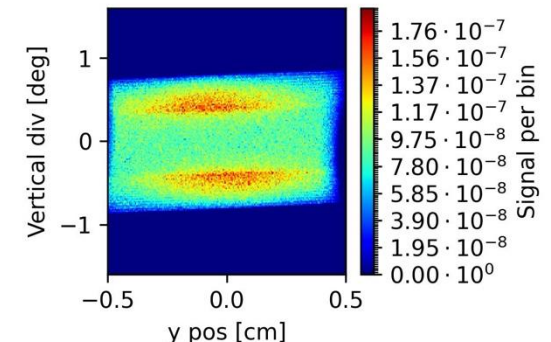
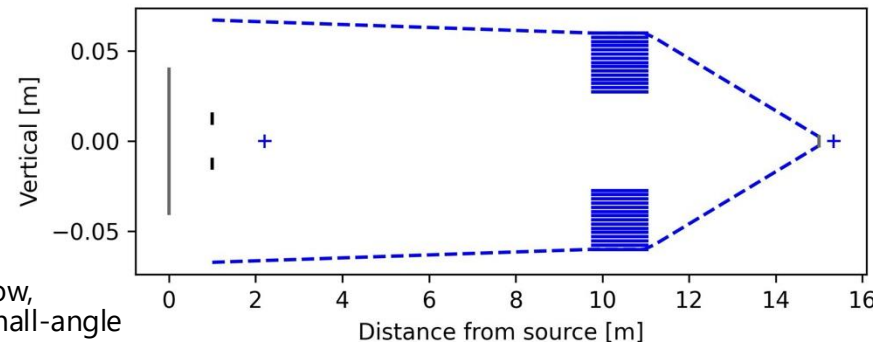
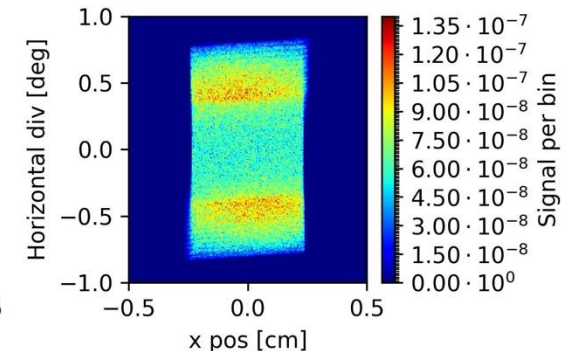
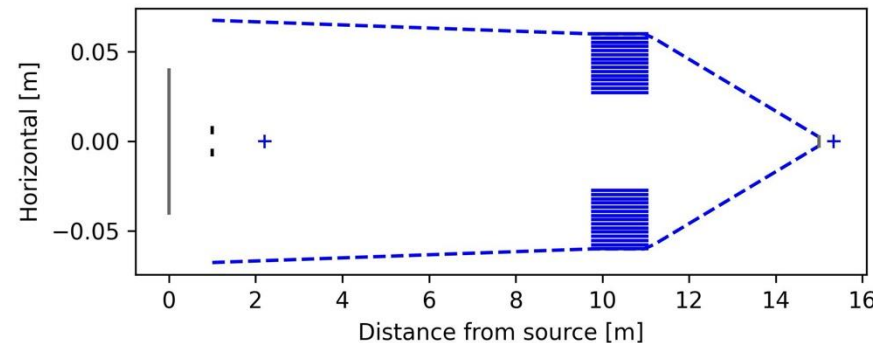
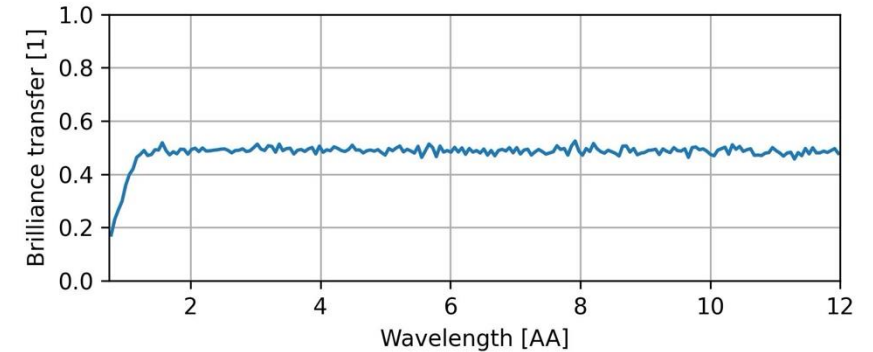
Solutions / Philosophies / Paradigms



Wolter optics

- Focusing design
- Allows large distance to sample
- Different kinds:
 - Point to point
 - Parallel beam to point
- Provides control over beam
- Limits
 - Gravity
 - Coating
 - Beamsize

$5 \times 5 \text{ mm}^2$
 $\pm 0.5^\circ \text{ H}$
 $\pm 0.8^\circ \text{ V}$



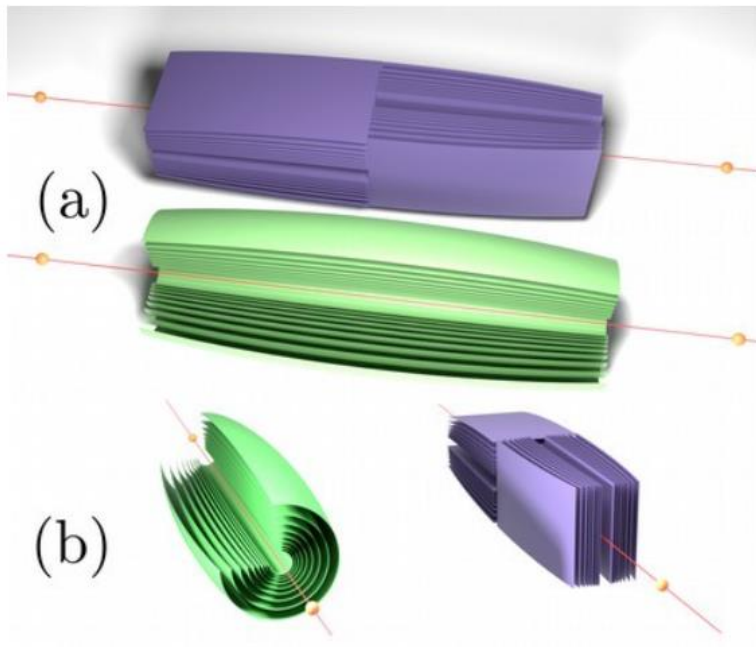
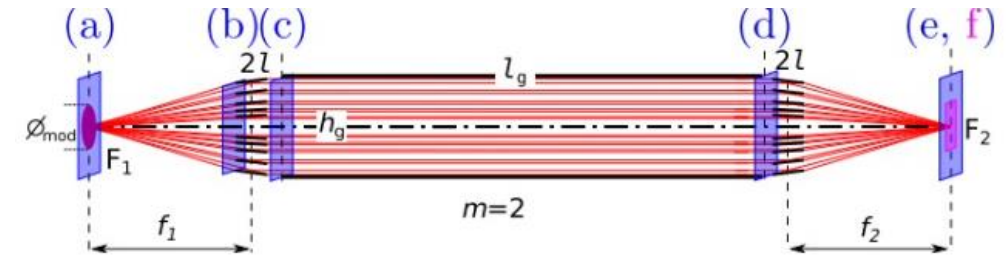
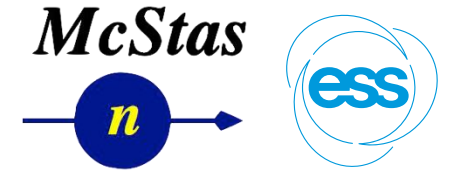
Simulation code by: Giacomo Resta

Dazhi Liu, Boris Khaykovich, Mikhail V. Gubarev, J. Lee Robertson, Lowell Crow, Brian D. Ramsey & David E. Moncton, Demonstration of a novel focusing small-angle neutron scattering instrument equipped with axisymmetric mirrors
Nature Communications 4, **2013**, Article number 2556

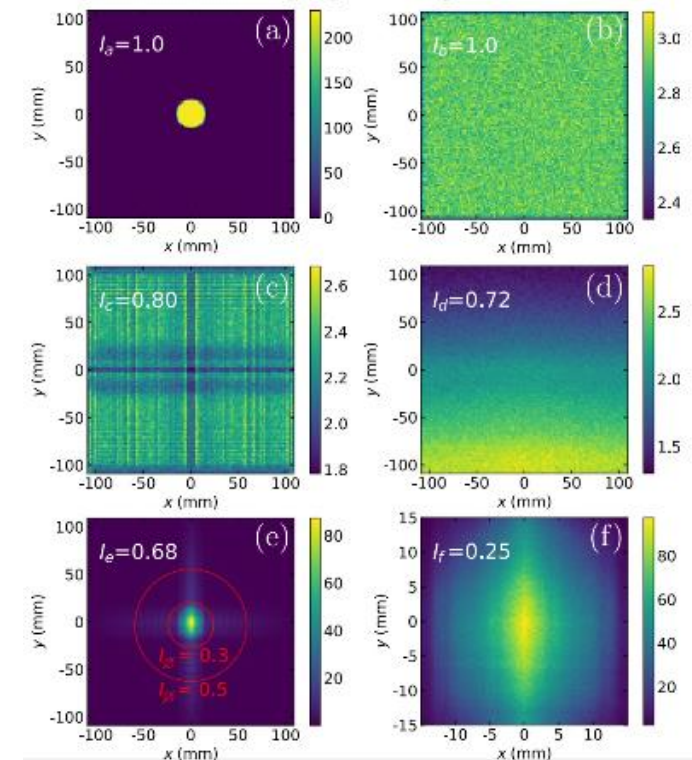
Solutions / Philosophies / Paradigms

Nested Mirror Optics

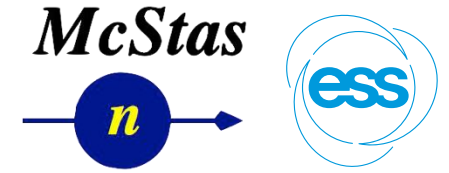
- Hybrid, focusing at either end
- Simulations here by Christoph Herb
- Contributed McStas component



including gravity



Solutions / Philosophies / Paradigms



Minimalist principle

- Calculate the required phase-space volume
 - Needed on sample + expected losses from gaps
- Only allow the guide to take in that amount of phase-space

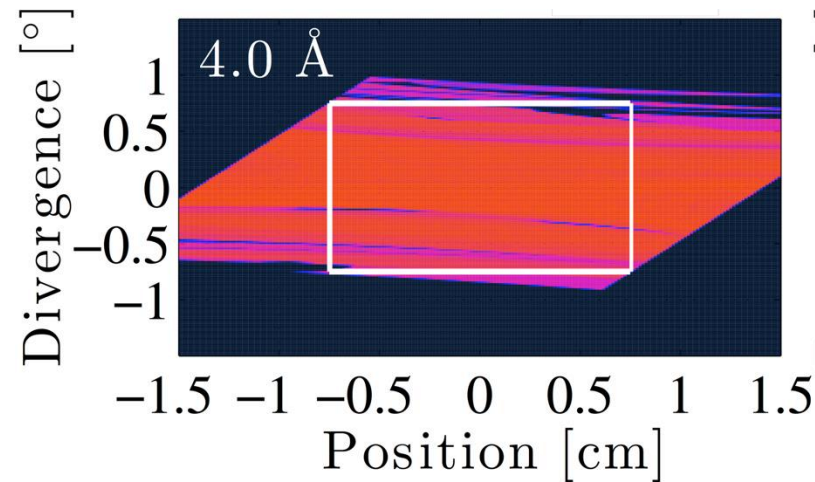
Focusing case

$$WM/L_M = 2\phi S$$

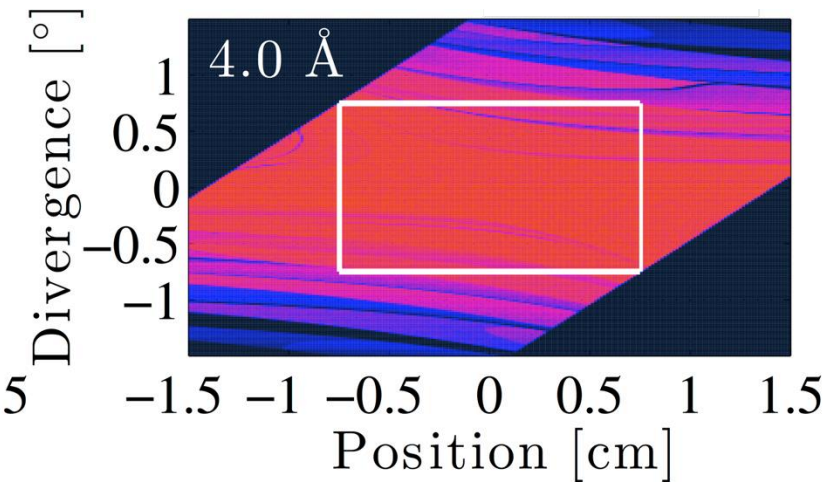
Non-focusing case

$$WM/L_M = 2\phi(S + 2L_S \tan(\phi))$$

Minimalist constraints



Brute force optimization



Solutions / Philosophies / Paradigms

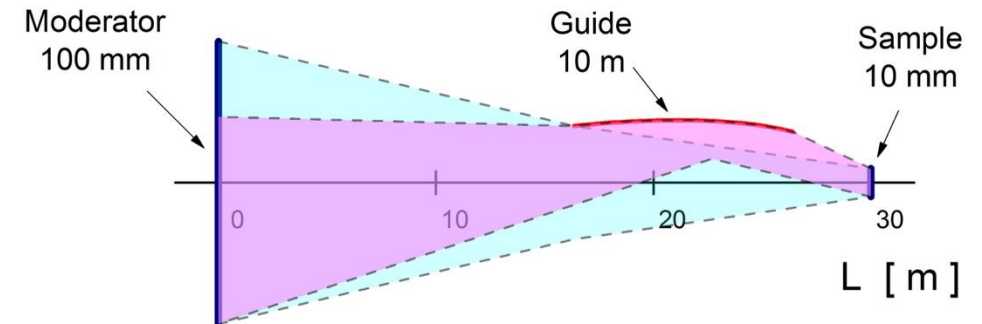
ORNL

- Shape developed by Ducu Stoica
- 3 Conditions
 - Direct line of sight
 - One reflection
 - No origins outside moderator

MATH* 

$$\left(\frac{y}{x} - \frac{\theta_0}{\beta} \right) x^\beta = C$$

Discover guide - horizontal



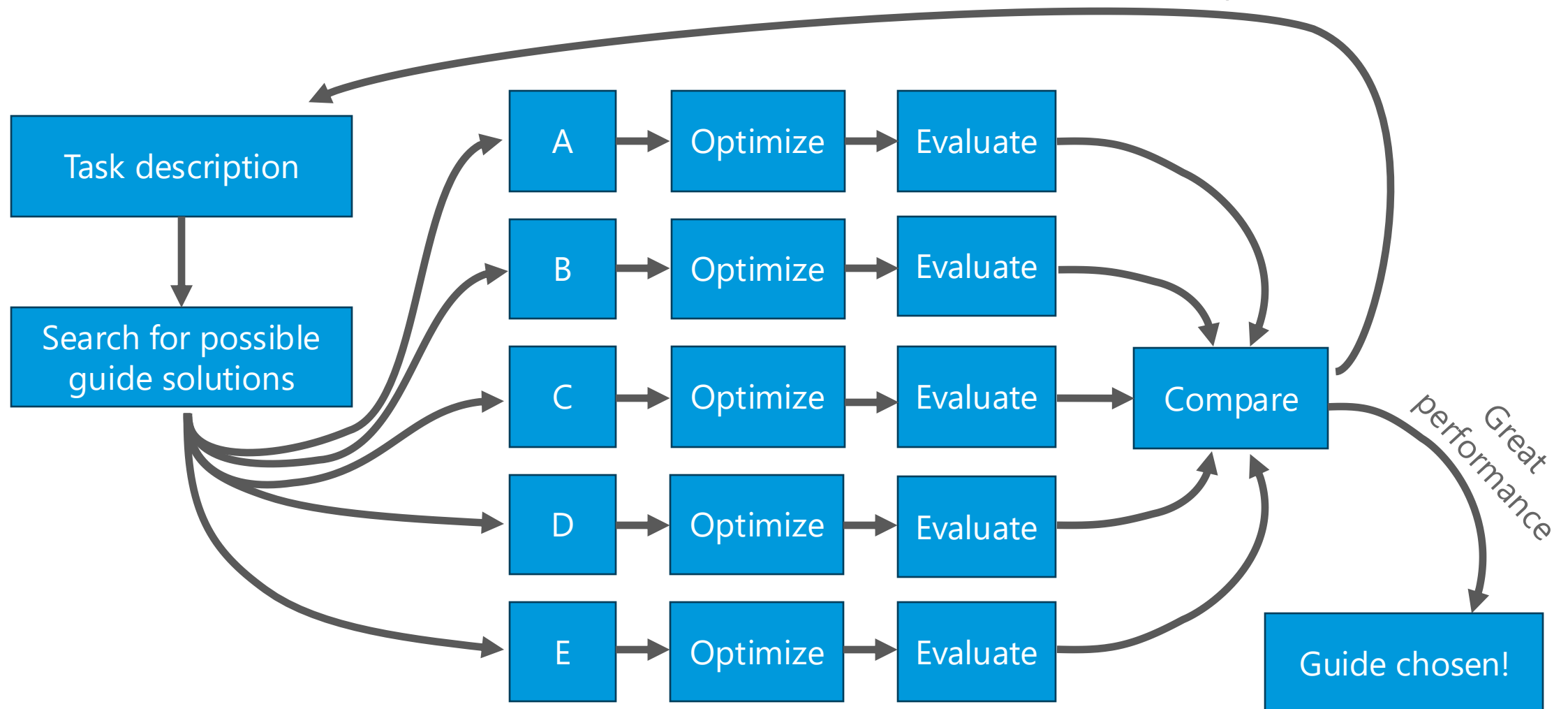
Every trajectory coming from the sample has to end on the moderator surface!

* Frost, M. J.; Stoica, A. D.; Huegle, T. A First-Principles Approach to the Optimization of Neutron-Focusing Guide Design and Development. *Journal of Neutron Research* **2019**, 20 (4), 131–135.

Designing a guide

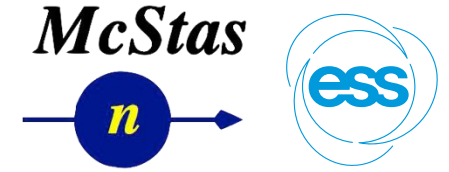
Workflow

Low brilliance transfer
Iterate task description



Guide simulation

Exercise



- Start from one of the two examples:
 - **Exercise_guides.instr** *Simpler instrument, easy to start with*
 - **Guide_BT_template.instr** *More advanced, has normalization to brilliance transfer*
- Main task: Explore how the guide parameters and geometry impact the results
- Extra tasks:
 - Scan guide length
 - Use Jupyter Notebook plot_brill to normalize data from Guide_BT_template
 - Introduce a gap by using two guide components
 - Use Guide_gravity and extend to 100 m length
 - Investigate the effect of gravity on the transport of long-wavelength neutrons



Thanks for your attention

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